

Exercise 3.I.2: Calculate the derivative of the deadweight loss measures (3.I.5) and (3.I.6) with respect to t . Show that, evaluated at $t = 0$, these derivatives are equal to zero but that if $h_1(p, u^0)$ is strictly decreasing in p_1 , they are strictly positive at all $t > 0$. Interpret.

Up to now, we have considered only the question of whether the consumer was better off at p^1 than at the initial price vector p^0 . We saw that both EV and CV provide a correct welfare ranking of p^0 and p^1 . Suppose, however, that p^0 is being compared with two possible price vectors p^1 and p^2 . In this case, p^1 is better than p^2 if and only if $EV(p^0, p^1, w) > EV(p^0, p^2, w)$, since

$$EV(p^0, p^1, w) - EV(p^0, p^2, w) = e(p^0, u^1) - e(p^0, u^2).$$

Thus, the EV measures $EV(p^0, p^1, w)$ and $EV(p^0, p^2, w)$ can be used not only to compare these two price vectors with p^0 but also to determine which of them is better for the consumer. A comparison of the compensating variations $CV(p^0, p^1, w)$ and $CV(p^0, p^2, w)$, however, will not necessarily rank p^1 and p^2 correctly. The problem is that the CV measure uses the new prices as the base prices in the money metric indirect utility function, using p^1 to calculate $CV(p^0, p^1, w)$ and p^2 to calculate $CV(p^0, p^2, w)$. So

$$CV(p^0, p^1, w) - CV(p^0, p^2, w) = e(p^2, u^0) - e(p^1, u^0),$$

which need not correctly rank p^1 and p^2 [see Exercise 3.I.4 and Chipman and Moore (1980)]. In other words, fixing p^0 , $EV(p^0, \cdot, w)$ is a valid indirect utility function (in fact, a money metric one), but $CV(p^0, \cdot, w)$ is not.²⁴

An interesting example of the comparison of several possible new price vectors arises when a government is considering which goods to tax. Suppose, for example, that two different taxes are being considered that could raise tax revenue of T : a tax on good 1 of t_1 (creating new price vector p^1) and a tax on good 2 of t_2 (creating new price vector p^2). Note that since they raise the same tax revenue, we have $t_1 x_1(p^1, w) = t_2 x_2(p^2, w) = T$ (see Figure 3.I.6). Because tax t_1

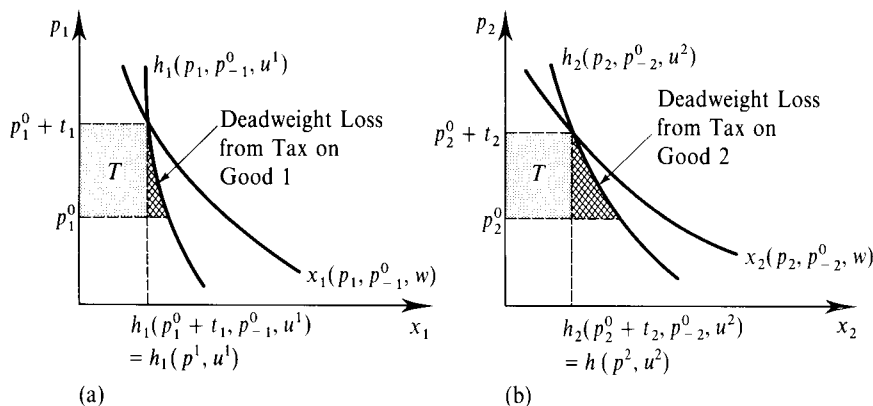


Figure 3.I.6

Comparing two taxes that raise revenue T .
(a) Tax on good 1.
(b) Tax on good 2.

is better than tax t_2 if and only if $EV(p^0, p^1, w) > EV(p^0, p^2, w)$, t_1 is better than t_2 if and only if $[(-T) - EV(p^0, p^1, w)] < [(-T) - EV(p^0, p^2, w)]$, that is, if and only if the deadweight loss arising under tax t_1 is less than that arising under tax t_2 .

24. Of course, we can rank p^1 and p^2 correctly by seeing whether $CV(p^1, p^2, w)$ is positive or negative.

In summary, if we know the consumer's expenditure function, we can precisely measure the welfare impact of a price change; moreover, we can do it in a convenient way (in dollars). In principle, this might well be the end of the story because, as we saw in Section 3.H, we can recover the consumer's preferences and expenditure function from the observable Walrasian demand function $x(p, w)$.²⁵ Before concluding, however, we consider two further issues. We first ask whether we may be able to say anything about the welfare effect of a price change when we *do not* have enough information to recover the consumer's expenditure function. We describe a test that provides a sufficient condition for the consumer's welfare to increase from the price change and that uses information only about the two price vectors p^0, p^1 and the initial consumption bundle $x(p^0, w)$. We then conclude by discussing in detail the extent to which the welfare change can be approximated by means of the area to the left of the market (Walrasian) demand curve, a topic of significant historical importance.

Welfare Analysis with Partial Information

In some circumstances, we may not be able to derive the consumer's expenditure function because we may have only limited information about her Walrasian demand function. Here we consider what can be said when the *only* information we possess is knowledge of the two price vectors p^0, p^1 and the consumer's initial consumption bundle $x^0 = x(p^0, w)$. We begin, in Proposition 3.I.1, by developing a simple sufficiency test for whether the consumer's welfare improves as a result of the price change.

Proposition 3.I.1: Suppose that the consumer has a locally nonsatiated rational preference relation $\succeq$. If $(p^1 - p^0) \cdot x^0 < 0$, then the consumer is strictly better off under price-wealth situation (p^1, w) than under (p^0, w) .

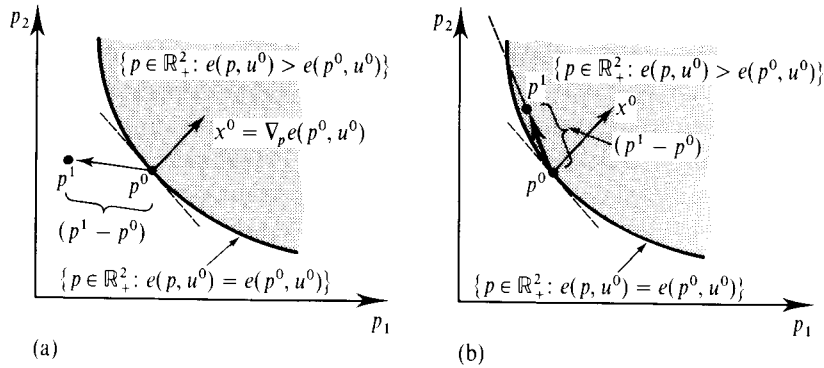
Proof: The result follows simply from revealed preference. Since $p^0 \cdot x^0 = w$ by Walras' law, if $(p^1 - p^0) \cdot x^0 < 0$, then $p^1 \cdot x^0 < w$. But if so, x^0 is still affordable under prices p^1 and is, moreover, in the interior of budget set $B_{p^1, w}$. By local nonsatiation, there must therefore be a consumption bundle in $B_{p^1, w}$ that the consumer strictly prefers to x^0 . ■

The test in Proposition 3.I.1 can be viewed as a first-order approximation to the true welfare change. To see this, take a first-order Taylor expansion of $e(p, u)$ around the initial prices p^0 :

$$e(p^1, u^0) = e(p^0, u^0) + (p^1 - p^0) \cdot \nabla_p e(p^0, u^0) + o(\|p^1 - p^0\|). \quad (3.I.7)$$

If $(p^1 - p^0) \cdot \nabla_p e(p^0, u^0) < 0$ and the second-order remainder term could be ignored, we would have $e(p^1, u^0) < e(p^0, u^0) = w$, and so we could conclude that the consumer's welfare is greater after the price change. But the concavity of $e(\cdot, u^0)$ in p implies that the remainder term is nonpositive. Therefore, ignoring the remainder term leads to no error here; we do have $e(p^1, u^0) < w$ if $(p^1 - p^0) \cdot \nabla_p e(p^0, u^0) < 0$. Using Proposition 3.G.1 then tells us that $(p^1 - p^0) \cdot \nabla_p e(p^0, u^0) = (p^1 - p^0) \cdot h(p^0, u^0) = (p^1 - p^0) \cdot x^0$, and so we get exactly the test in Proposition 3.I.1.

25. As a practical matter, in applications you should use whatever are the state-of-the-art techniques for performing this recovery.

**Figure 3.1.7**

The welfare test of Propositions 3.1.1 and 3.1.2.

(a) $(p^1 - p^0) \cdot x^0 < 0$
 (b) $(p^1 - p^0) \cdot x^0 > 0$

What if $(p^1 - p^0) \cdot x^0 > 0$? Can we then say anything about the direction of change in welfare? As a general matter, no. However, examination of the first-order Taylor expansion (3.1.7) tells us that we get a definite conclusion if the price change is, in an appropriate sense, small enough because the remainder term then becomes insignificant relative to the first-order term and can be neglected. This gives the result shown in Proposition 3.1.2.

Proposition 3.1.2: Suppose that the consumer has a differentiable expenditure function. Then if $(p^1 - p^0) \cdot x^0 > 0$, there is a sufficiently small $\bar{\alpha} \in (0, 1)$ such that for all $\alpha < \bar{\alpha}$, we have $e((1 - \alpha)p^0 + \alpha p^1, u^0) > w$, and so the consumer is strictly better off under price-wealth situation (p^0, w) than under $((1 - \alpha)p^0 + \alpha p^1, w)$.

Figure 3.1.7 illustrates these results for the cases where p^1 is such that $(p^1 - p^0) \cdot x^0 < 0$ [panel (a)] and $(p^1 - p^0) \cdot x^0 > 0$ [panel (b)]. In the figure the set of prices $\{p \in \mathbb{R}_+^2 : e(p, u^0) \geq e(p^0, u^0)\}$ is drawn in price space. The concavity of $e(\cdot, u)$ gives it the shape depicted. The initial price vector p^0 lies in this set. By Proposition 3.G.1, the gradient of the expenditure function at this point, $\nabla_p e(p^0, u^0)$, is equal to x^0 , the initial consumption bundle. The vector $(p^1 - p^0)$ is the vector connecting point p^0 to the new price point p^1 . Figure 3.1.7(a) shows a case where $(p^1 - p^0) \cdot x^0 < 0$. As can be seen there, p^1 lies outside of the set $\{p \in \mathbb{R}_+^2 : e(p, u^0) \geq e(p^0, u^0)\}$, and so we must have $e(p^0, u^0) > e(p^1, u^0)$. In Figure 3.1.7(b), on the other hand, we show a case where $(p^1 - p^0) \cdot x^0 > 0$. Proposition 3.1.2 can be interpreted as asserting that in this case if $(p^1 - p^0)$ is small enough, then $e(p^0, u^0) < e(p^1, u^0)$. This can be seen in Figure 3.1.7(b), because if $(p^1 - p^0) \cdot x^0 > 0$ and p^1 is close enough to p^0 [in the ray with direction $p^1 - p^0$], then price vector p^1 lies in the set $\{p \in \mathbb{R}_+^2 : e(p, u^0) > e(p^0, u^0)\}$.

Using the Area to the Left of the Walrasian (Market) Demand Curve as an Approximate Welfare Measure

Improvements in computational abilities have made the recovery of the consumer's preferences/expenditure function from observed demand behavior, along the lines discussed in Section 3.1, far easier than was previously the case.²⁶ Traditionally,

26. They have also made it much easier to estimate complicated demand systems that are explicitly derived from utility maximization and from which the parameters of the expenditure function can be derived directly.

however, it has been common practice in applied analyses to rely on approximations of the true welfare change.

We have already seen in (3.I.3) and (3.I.4) that the welfare change induced by a change in the price of good 1 can be exactly computed by using the area to the left of an appropriate Hicksian demand curve. However, these measures present the problem of not being directly observable. A simpler procedure that has seen extensive use appeals to the Walrasian (market) demand curve instead. We call this estimate of welfare change the *area variation* measure (or AV):

$$AV(p^0, p^1, w) = \int_{p_1^1}^{p_1^0} x_1(p_1, \bar{p}_{-1}, w) dp_1. \quad (3.I.8)$$

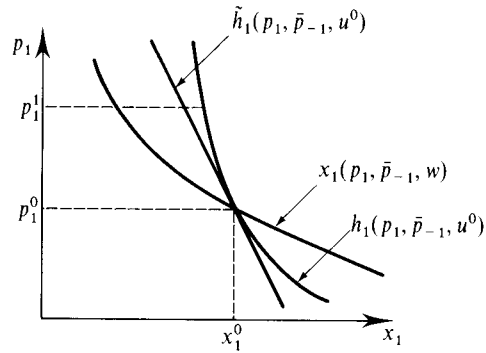
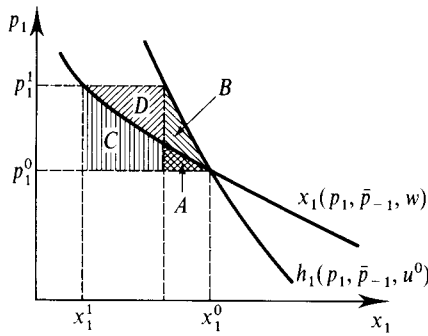
If there are no wealth effects for good 1, then, as we have discussed, $x_1(p, w) = h_1(p, u^0) = h_1(p, u^1)$ for all p and the area variation measure is exactly equal to the equivalent and compensating variation measures. This corresponds to the case studied by Marshall (1920) in which the marginal utility of numeraire is constant. In this circumstance, where the AV measure gives an exact measure of welfare change, the measure is known as the change in *Marshallian consumer surplus*.

More generally, as Figures 3.I.3(a) and 3.I.3(b) make clear, when good 1 is a normal good, the area variation measure overstates the compensating variation and understates the equivalent variation (convince yourself that this is true both when p_1 falls and when p_1 rises). When good 1 is inferior, the reverse relations hold. Thus, when evaluating the welfare change from a change in prices of several goods, or when comparing two different possible price changes, the area variation measure need not give a correct evaluation of welfare change (e.g., see Exercise 3.I.10).

Naturally enough, however, if the wealth effects for the goods under consideration are small, the approximation errors are also small and the area variation measure is almost correct. Marshall argued that if a good is just one commodity among many, then because one extra unit of wealth will spread itself around, the wealth effect for the commodity is bound to be small; therefore, no significant errors will be made by evaluating the welfare effects of price changes for that good using the area measure. This idea can be made precise; for an advanced treatment, see Vives (1987). It is important, however, not to fall into the fallacy of composition; if we deal with a large number of commodities, then while the approximating error may be small for each individually, it may nevertheless not be small in the aggregate.

If $(p_1^1 - p_1^0)$ is small, then the error involved using the area variation measure becomes small as a fraction of the true welfare change. Consider, for example, the compensating variation.²⁷ In Figure 3.I.8, we see that the area $B + D$, which measures the difference between the area variation and the true compensating variation, becomes small as a fraction of the true compensating variation when $(p_1^1 - p_1^0)$ is small. This might seem to suggest that the area variation measure is a good approximation of the compensating variation measure for small price changes. Note, however, that the same property would hold if instead of the Walrasian demand

27. All the points that follow apply to the equivalent variation as well.

**Figure 3.I.8 (left)**

The error in using the area variation measure of welfare change.

Figure 3.I.9 (right)

A first-order approximation of $h(p, u^0)$ at p^0 .

function we were to use *any* function that takes the value $x_1(p_1^0, p_{-1}^0, w)$ at p_1^0 .²⁸ In fact, the approximation error may be quite large as a fraction of the deadweight loss [this point is emphasized by Hausman (1981)]. In Figure 3.I.8, for example, the deadweight loss calculated using the Walrasian demand curve is the area $A + C$, whereas the real one is the area $A + B$. The percentage difference between these two areas need not grow small as the price change grows small.²⁹

When $(p_1^1 - p_1^0)$ is small, there is a superior approximation procedure available. In particular, suppose we take a first-order Taylor approximation of $h(p, u^0)$ at p^0

$$\tilde{h}(p, u^0) = h(p^0, u^0) + D_p h(p^0, u^0)(p - p^0)$$

and we calculate

$$\int_{p_1^1}^{p_1^0} \tilde{h}_1(p_1, \bar{p}_{-1}, u^0) dp_1 \quad (3.I.9)$$

as our approximation of the welfare change. The function $\tilde{h}_1(p_1, \bar{p}_{-1}, u^0)$ is depicted in Figure 3.I.9. As can be seen in the figure, because $\tilde{h}_1(p_1, \bar{p}_{-1}, u^0)$ has the same slope as the true Hicksian demand function $h_1(p, u^0)$ at p^0 , for small price changes this approximation comes closer than expression (3.I.8) to the true welfare change (and in contrast with the area variation measure, it provides an adequate approximation to the deadweight loss). Because the Hicksian demand curve is the first derivative of the expenditure function, this first-order expansion of the Hicksian demand function at p^0 is, in essence, a second-order expansion of the expenditure function around p^0 . Thus, this approximation can be viewed as the natural extension of the first-order test discussed above; see expression (3.I.7).

The approximation in (3.I.9) is directly computable from knowledge of the observable Walrasian demand function $x_1(p, w)$. To see this, note that because $h(p^0, u^0) = x(p^0, w)$ and $D_p h(p^0, u^0) = S(p^0, w)$, $\tilde{h}(p, u^0)$ can be expressed solely in terms that involve the Walrasian demand function and its derivatives at the point

28. In effect, the property identified here amounts to saying that the Walrasian demand function provides a first-order approximation to the compensating variation. Indeed, note that the derivatives of $CV(p^1, p^0, w)$, $EV(p^1, p^0, w)$, and $AV(p^1, p^0, w)$ with respect to p_1^1 evaluated at p_1^0 are all precisely $x_1(p_1^0, p_{-1}^0, w)$.

29. Thus, for example, in the problem discussed above where we compare the deadweight losses induced by taxes on two different commodities that both raise revenue T , the area variation measure need not give the correct ranking even for small taxes.

(p^0, w) :

$$\tilde{h}(p, u^0) = x(p^0, w) + S(p^0, w)(p - p^0).$$

In particular, since only the price of good 1 is changing, we have

$$\tilde{h}_1(p_1, \bar{p}_{-1}, u^0) = x_1(p_1^0, \bar{p}_{-1}, w) + s_{11}(p_1^0, \bar{p}_{-1}, w)(p_1 - p_1^0),$$

where

$$s_{11}(p_1^0, \bar{p}_{-1}, w) = \frac{\partial x_1(p^0, w)}{\partial p_1} + \frac{\partial x_1(p^0, w)}{\partial w} x_1(p^0, w).$$

When $(p^1 - p^0)$ is small, this procedure provides a better approximation to the true compensating variation than does the area variation measure. However, if $(p^1 - p^0)$ is large, we cannot tell which is the better approximation. It is entirely possible for the area variation measure to be superior. After all, its use guarantees some sensitivity of the approximation to demand behavior away from p^0 , whereas the use of $\tilde{h}(p, u^0)$ does not.

3.J The Strong Axiom of Revealed Preference

We have seen that in the context of consumer demand theory, consumer choice may satisfy the weak axiom but not be capable of being generated by a rational preference relation (see Sections 2.F and 3.G). We could therefore ask: Can we find a necessary and sufficient consistency condition on consumer demand behavior that is in the same style as the WA but that does imply that demand behavior can be rationalized by preferences? The answer is “yes”, and it was provided by Houthakker (1950) in the form of the *strong axiom of revealed preference* (SA), a kind of recursive closure of the weak axiom.³⁰

Definition 3.J.1: The market demand function $x(p, w)$ satisfies the *strong axiom of revealed preference* (the SA) if for any list

$$(p^1, w^1), \dots, (p^N, w^N)$$

with $x(p^{n+1}, w^{n+1}) \neq x(p^n, w^n)$ for all $n \leq N-1$, we have $p^N \cdot x(p^1, w^1) > w^N$ whenever $p^n \cdot x(p^{n+1}, w^{n+1}) \leq w^n$ for all $n \leq N-1$.

In words, if $x(p^1, w^1)$ is *directly or indirectly revealed preferred* to $x(p^N, w^N)$, then $x(p^N, w^N)$ cannot be (directly) revealed preferred to $x(p^1, w^1)$ [so $x(p^1, w^1)$ cannot be affordable at (p^N, w^N)]. For example, the SA was violated in Example 2.F.1. It is clear that the SA is satisfied if demand originates in rational preferences. The converse is a deeper result. It is stated in Proposition 3.J.1; the proof, which is advanced, is presented in small type.

Proposition 3.J.1: If the Walrasian demand function $x(p, w)$ satisfies the strong axiom of revealed preference then there is a rational preference relation $\succsim$ that rationalizes $x(p, w)$, that is, such that for all (p, w) , $x(p, w) \succ y$ for every $y \neq x(p, w)$ with $y \in B_{p, w}$.

30. For an informal account of revealed preference theory after Samuelson, see Mas-Colell (1982).

Proof: We follow Richter (1966). His proof is based on set theory and differs markedly from the differential equations techniques used originally by Houthakker.³¹

Define a relation $\succ^1$ on commodity vectors by letting $x \succ^1 y$ whenever $x \neq y$ and we have $x = x(p, w)$ and $p \cdot y \leq w$ for some (p, w) . The relation $\succ^1$ can be read as “directly revealed preferred to.” From $\succ^1$ define a new relation $\succ^2$, to be read as “directly or indirectly revealed preferred to,” by letting $x \succ^2 y$ whenever there is a chain $x^1 \succ^1 x^2 \succ^1 \dots \succ^1 x^N$ with $x^1 = x$ and $x^N = y$. Observe that, by construction, $\succ^2$ is transitive. According to the SA, $\succ^2$ is also irreflexive (i.e., $x \succ^2 x$ is impossible). A certain axiom of set theory (known as Zorn’s lemma) tells us the following: *Every relation $\succ^2$ that is transitive and irreflexive (called a partial order) has a total extension $\succ^3$, an irreflexive and transitive relation such that, first, $x \succ^2 y$ implies $x \succ^3 y$ and, second, whenever $x \neq y$, we have either $x \succ^3 y$ or $y \succ^3 x$.* Finally, we can define $\succ$ by letting $x \succ y$ whenever $x = y$ or $x \succ^3 y$. It is not difficult now to verify that $\succ$ is complete and transitive and that $x(p, w) \succ y$ whenever $p \cdot y \leq w$ and $y \neq x(p, w)$. ■

The proof of Proposition 3.J.1 uses only the single-valuedness of $x(p, w)$. Provided choice is single-valued, the same result applies to the abstract theory of choice of Chapter 1. The fact that the budgets are competitive is immaterial.

In Exercise 3.J.1, you are asked to show that the WA is equivalent to the SA when $L = 2$. Hence, by Proposition 3.J.1, when $L = 2$ and demand satisfies the WA, we can always find a rationalizing preference relation, a result that we have already seen in Section 3.H. When $L > 2$, however, the SA is stronger than the WA. In fact, Proposition 3.J.1 tells us that a choice-based theory of demand founded on the strong axiom is essentially equivalent to the preference-based theory of demand presented in this chapter.

The strong axiom is therefore essentially equivalent both to the rational preference hypothesis and to the symmetry and negative semidefiniteness of the Slutsky matrix. We have seen that the weak axiom is essentially equivalent to the negative semidefiniteness of the Slutsky matrix. It is therefore natural to ask whether there is an assumption on preferences that is weaker than rationality and that leads to a theory of consumer demand equivalent to that based on the WA. Violations of the SA mean cycling choice, and violations of the symmetry of the Slutsky matrix generate path dependence in attempts to “integrate back” to preferences. This suggests preferences that may violate the transitivity axiom. See the appendix with W. Shafer in Kihlstrom, Mas-Colell, and Sonnenschein (1976) for further discussion of this point.

APPENDIX A: CONTINUITY AND DIFFERENTIABILITY PROPERTIES OF WALRASIAN DEMAND

In this appendix, we investigate the continuity and differentiability properties of the Walrasian demand correspondence $x(p, w)$. We assume that $x \gg 0$ for all $(p, w) \gg 0$ and $x \in x(p, w)$.

31. Yet a third approach, based on linear programming techniques, was provided by Afriat (1967).

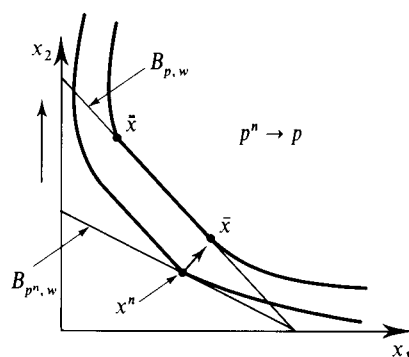


Figure 3.AA.1
An upper
hemicontinuous
Walrasian demand
correspondence.

Continuity

Because $x(p, w)$ is, in general, a correspondence, we begin by introducing a generalization of the more familiar continuity property for functions, called *upper hemicontinuity*.

Definition 3.AA.1: The Walrasian demand correspondence $x(p, w)$ is *upper hemicontinuous* at $(\bar{p}, \bar{w})$ if whenever $(p^n, w^n) \rightarrow (\bar{p}, \bar{w})$, $x^n \in x(p^n, w^n)$ for all n , and $x = \lim_{n \rightarrow \infty} x^n$, we have $x \in x(p, w)$.³²

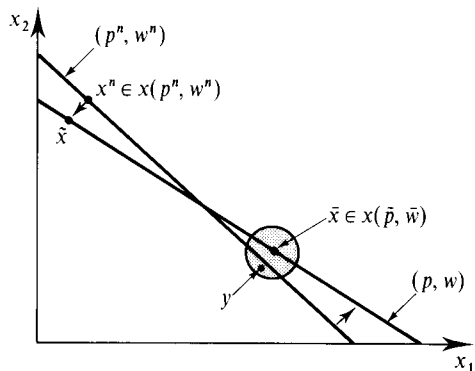
In words, a demand correspondence is upper hemicontinuous at $(\bar{p}, \bar{w})$ if for any sequence of price–wealth pairs the limit of any sequence of optimal demand bundles is optimal (although not necessarily uniquely so) at the limiting price–wealth pair. If $x(p, w)$ is single-valued at all $(p, w) \gg 0$, this notion is equivalent to the usual continuity property for functions.

Figure 3.AA.1 depicts an upper hemicontinuous demand correspondence: When $p^n \rightarrow p$, $x(\cdot, w)$ exhibits a jump in demand behavior at the price vector p , being x^n for all p^n but suddenly becoming the interval of consumption bundles $[\bar{x}, \bar{x}]$ at p . It is upper hemicontinuous because $\bar{x}$ (the limiting optimum for p^n along the sequence) is an element of segment $[\bar{x}, \bar{x}]$ (the set of optima at price vector p). See Section M.H of the Mathematical Appendix for further details on upper hemicontinuity.

Proposition 3.AA.1: Suppose that $u(\cdot)$ is a continuous utility function representing locally nonsatiated preferences $\succsim$ on the consumption set $X = \mathbb{R}_+^L$. Then the derived demand correspondence $x(p, w)$ is upper hemicontinuous at all $(p, w) \gg 0$. Moreover, if $x(p, w)$ is a function [i.e., if $x(p, w)$ has a single element for all (p, w)], then it is continuous at all $(p, w) \gg 0$.

Proof: To verify upper hemicontinuity, suppose that we had a sequence $\{(p^n, w^n)\}_{n=1}^{\infty} \rightarrow (\bar{p}, \bar{w}) \gg 0$ and a sequence $\{x^n\}_{n=1}^{\infty}$ with $x^n \in x(p^n, w^n)$ for all n , such that $x^n \rightarrow \bar{x}$ and $\bar{x} \notin x(\bar{p}, \bar{w})$. Because $p^n \cdot x^n \leq w^n$ for all n , taking limits as $n \rightarrow \infty$, we conclude that $\bar{p} \cdot \bar{x} \leq \bar{w}$. Thus, $\bar{x}$ is a feasible consumption bundle when the budget set is $B_{\bar{p}, \bar{w}}$. However, since it is not optimal in this set, it must be that $u(\bar{x}) > u(\bar{x})$ for some $\bar{x} \in B_{\bar{p}, \bar{w}}$.

32. We use the notation $z^n \rightarrow z$ as synonymous with $z = \lim_{n \rightarrow \infty} z^n$. This definition of upper hemicontinuity applies only to correspondences that are “locally bounded” (see Section M.H of the Mathematical Appendix). Under our assumptions, the Walrasian demand correspondence satisfies this property at all $(p, w) \gg 0$.



By the continuity of $u(\cdot)$, there is a y arbitrarily close to $\bar{x}$ such that $p \cdot y < w$ and $u(y) > u(\bar{x})$. This bundle y is illustrated in Figure 3.AA.2.

Note that if n is large enough, we will have $p^n \cdot y < w^n$ [since $(p^n, w^n) \rightarrow (p, w)$]. Hence, y is an element of the budget set B_{p^n, w^n} , and we must have $u(x^n) \geq u(y)$ because $x^n \in x(p^n, w^n)$. Taking limits as $n \rightarrow \infty$, the continuity of $u(\cdot)$ then implies that $u(\bar{x}) \geq u(y)$, which gives us a contradiction. We must therefore have $\bar{x} \in x(p, w)$, establishing upper hemicontinuity of $x(p, w)$.

The same argument also establishes continuity if $x(p, w)$ is in fact a function. ■

Suppose that the consumption set is an arbitrary closed set $X \subset \mathbb{R}_+^L$. Then the continuity (or upper hemicontinuity) property still follows at any $(\bar{p}, \bar{w})$ that passes the following (*locally cheaper consumption*) test: "Suppose that $x \in X$ is affordable (i.e., $\bar{p} \cdot x \leq \bar{w}$). Then there is a $y \in X$ arbitrarily close to x and that costs less than $\bar{w}$ (i.e., $\bar{p} \cdot y < \bar{w}$)." For example, in Figure 3.AA.3, commodity 2 is available only in indivisible unit amounts. The locally cheaper test then fails at the price-wealth point $(\bar{p}, \bar{w}) = (1, \bar{w}, \bar{w})$, where a unit of good 2 becomes just affordable. You can easily verify by examining the figure [in which the dashed line indicates indifference between the points $(0, 1)$ and z] that demand will fail to be upper hemicontinuous when $p_2 = \bar{w}$. In particular, for price-wealth points $(p^n, \bar{w})$ such that $p_1^n = 1$ and $p_2^n > \bar{w}$, $x(p^n, \bar{w})$ involves only the consumption of good 1; whereas at $(\bar{p}, \bar{w}) = (1, \bar{w}, \bar{w})$, we have $x(\bar{p}, \bar{w}) = (0, 1)$. Note that the proof of Proposition 3.AA.1 fails when the locally cheaper consumption condition does not hold because we cannot find a consumption bundle y with the properties described there.

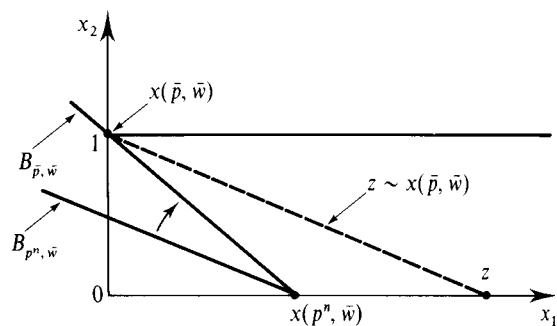


Figure 3.AA.2 (left)
Finding a bundle y such that $p \cdot y < w$ and $u(y) > u(\bar{x})$.

Figure 3.AA.3 (right)
The locally cheaper test fails at price-wealth pair $(\bar{p}, \bar{w}) = (1, \bar{w}, \bar{w})$.

Differentiability

Proposition 3.AA.1 has established that if $x(p, w)$ is a function, then it is continuous. Often it is convenient that it be differentiable as well. We now discuss when this is so. We assume for the remaining paragraphs that $u(\cdot)$ is strictly quasiconcave and twice continuously differentiable and that $\nabla u(x) \neq 0$ for all x .

As we have shown in Section 3.D, the first-order conditions for the UMP imply that $x(p, w) \gg 0$ is, for some $\lambda > 0$, the unique solution of the system of $L + 1$ equations in $L + 1$ unknowns:

$$\begin{aligned}\nabla u(x) - \lambda p &= 0 \\ p \cdot x - w &= 0.\end{aligned}$$

Therefore, the *implicit function theorem* (see Section M.E of the Mathematical Appendix) tells us that the differentiability of the solution $x(p, w)$ as a function of the parameters (p, w) of the system depends on the Jacobian matrix of this system having a nonzero determinant. The Jacobian matrix [i.e., the derivative matrix of the $L + 1$ component functions with respect to the $L + 1$ variables (x, λ)] is

$$\begin{bmatrix} D^2u(x) & -p \\ p^T & 0 \end{bmatrix}.$$

Since $\nabla u(x) = \lambda p$ and $\lambda > 0$, the determinant of this matrix is nonzero if and only if the determinant of the *bordered Hessian* of $u(x)$ at x is nonzero:

$$\begin{vmatrix} D^2u(x) & \nabla u(x) \\ [\nabla u(x)]^T & 0 \end{vmatrix} \neq 0.$$

This condition has a straightforward geometric interpretation. It means that the indifference set through x has a nonzero curvature at x ; it is not (even infinitesimally) flat. This condition is a slight technical strengthening of strict quasiconcavity [just as the strictly concave function $f(x) = -(x^4)$ has $f''(0) = 0$, a strictly quasiconcave function could have a bordered Hessian determinant that is zero at a point].

We conclude, therefore, that $x(p, w)$ is differentiable *if and only if* the determinant of the bordered Hessian of $u(\cdot)$ is nonzero at $x(p, w)$. It is worth noting the following interesting fact (which we shall not prove here): If $x(p, w)$ is differentiable at (p, w) , then the Slutsky matrix $S(p, w)$ has maximal possible rank; that is, the rank of $S(p, w)$ equals $L - 1$.³³

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33. This statement applies only to demand generated from a twice continuously differentiable utility function. It need not be true when this condition is not met. For example, the demand function $x(p, w) = (w/(p_1 + p_2), w/(p_1 + p_2))$ is differentiable, and it is generated by the utility function $u(x) = \text{Min} \{x_1, x_2\}$, which is not twice continuously differentiable at all x . The substitution matrix for this demand function has all its entries equal to zero and therefore has rank equal to zero.

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EXERCISES

3.B.1^A In text.

3.B.2^B The preference relation $\succsim$ defined on the consumption set $X = \mathbb{R}_+^L$ is said to be *weakly monotone* if and only if $x \geq y$ implies that $x \succsim y$. Show that if $\succsim$ is transitive, locally nonsatiated, and weakly monotone, then it is monotone.

3.B.3^A Draw a convex preference relation that is locally nonsatiated but is not monotone.

3.C.1^B Verify that the lexicographic ordering is complete, transitive, strongly monotone, and strictly convex.

3.C.2^B Show that if $u(\cdot)$ is a continuous utility function representing $\succsim$, then $\succsim$ is continuous.

3.C.3^C Show that if for every x the upper and lower contour sets $\{y \in \mathbb{R}_+^L : y \succsim x\}$ and $\{y \in \mathbb{R}_+^L : x \succ y\}$ are closed, then $\succsim$ is continuous according to Definition 3.C.1.

3.C.4^B Exhibit an example of a preference relation that is not continuous but is representable by a utility function.

3.C.5^C Establish the following two results:

(a) A continuous $\succsim$ is homothetic if and only if it admits a utility function $u(x)$ that is homogeneous of degree one; i.e., $u(\alpha x) = \alpha u(x)$ for all $\alpha > 0$.

(b) A continuous $\succsim$ on $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ is quasilinear with respect to the first commodity if and only if it admits a utility function $u(x)$ of the form $u(x) = x_1 + \phi(x_2, \dots, x_L)$. [Hint: The existence of some continuous utility representation is guaranteed by Proposition 3.G.1.]

After answering (a) and (b), argue that these properties of $u(\cdot)$ are cardinal.

3.C.6^B Suppose that in a two-commodity world, the consumer's utility function takes the form $u(x) = [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho}$. This utility function is known as the *constant elasticity of substitution* (or CES) utility function.

(a) Show that when $\rho = 1$, indifference curves become linear.

(b) Show that as $\rho \rightarrow 0$, this utility function comes to represent the same preferences as the (generalized) Cobb–Douglas utility function $u(x) = x_1^{\alpha_1} x_2^{\alpha_2}$.

(c) Show that as $\rho \rightarrow -\infty$, indifference curves become “right angles”; that is, this utility function has in the limit the indifference map of the Leontief utility function $u(x_1, x_2) = \text{Min}\{x_1, x_2\}$.

3.D.1^A In text.

3.D.2^A In text.

3.D.3^B Suppose that $u(x)$ is differentiable and strictly quasiconcave and that the Walrasian demand function $x(p, w)$ is differentiable. Show the following:

(a) If $u(x)$ is homogeneous of degree one, then the Walrasian demand function $x(p, w)$ and the indirect utility function $v(p, w)$ are homogeneous of degree one [and hence can be written in the form $x(p, w) = w\tilde{x}(p)$ and $v(p, w) = w\tilde{v}(p)$] and the wealth expansion path (see Section 2.E) is a straight line through the origin. What does this imply about the wealth elasticities of demand?

(b) If $u(x)$ is strictly quasiconcave and $v(p, w)$ is homogeneous of degree one in w , then $u(x)$ must be homogeneous of degree one.

3.D.4^B Let $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ denote the consumption set, and assume that preferences are strictly convex and quasilinear. Normalize $p_1 = 1$.

(a) Show that the Walrasian demand functions for goods $2, \dots, L$ are independent of wealth. What does this imply about the wealth effect (see Section 2.E) of demand for good 1?

(b) Argue that the indirect utility function can be written in the form $v(p, w) = w + \phi(p)$ for some function $\phi(\cdot)$.

(c) Suppose, for simplicity, that $L = 2$, and write the consumer's utility function as $u(x_1, x_2) = x_1 + \eta(x_2)$. Now, however, let the consumption set be $\mathbb{R}_+^2$ so that there is a nonnegativity constraint on consumption of the numeraire x_1 . Fix prices p , and examine how the consumer's Walrasian demand changes as wealth w varies. When is the nonnegativity constraint on the numeraire irrelevant?

3.D.5^B Consider again the CES utility function of Exercise 3.C.6, and assume that $\alpha_1 = \alpha_2 = 1$.

(a) Compute the Walrasian demand and indirect utility functions for this utility function.

(b) Verify that these two functions satisfy all the properties of Propositions 3.D.2 and 3.D.3.

(c) Derive the Walrasian demand correspondence and indirect utility function for the case of linear utility and the case of Leontief utility (see Exercise 3.C.6). Show that the CES Walrasian demand and indirect utility functions approach these as ρ approaches 1 and $-\infty$, respectively.

(d) The *elasticity of substitution between goods 1 and 2* is defined as

$$\xi_{12}(p, w) = - \frac{\partial[x_1(p, w)/x_2(p, w)]}{\partial[p_1/p_2]} \frac{p_1/p_2}{x_1(p, w)/x_2(p, w)}.$$

Show that for the CES utility function, $\xi_{12}(p, w) = 1/(1 - \rho)$, thus justifying its name. What is $\xi_{12}(p, w)$ for the linear, Leontief, and Cobb–Douglas utility functions?

3.D.6^B Consider the three-good setting in which the consumer has utility function $u(x) = (x_1 - b_1)^\alpha (x_2 - b_2)^\beta (x_3 - b_3)^\gamma$.

(a) Why can you assume that $\alpha + \beta + \gamma = 1$ without loss of generality? Do so for the rest of the problem.

(b) Write down the first-order conditions for the UMP, and derive the consumer's Walrasian demand and indirect utility functions. This system of demands is known as the *linear expenditure system* and is due to Stone (1954).

(c) Verify that these demand functions satisfy the properties listed in Propositions 3.D.2 and 3.D.3.

3.D.7^B There are two commodities. We are given two budget sets B_{p^0, w^0} and B_{p^1, w^1} described, respectively, by $p^0 = (1, 1)$, $w^0 = 8$ and $p^1 = (1, 4)$, $w^1 = 26$. The observed choice at (p^0, w^0) is $x^0 = (4, 4)$. At (p^1, w^1) , we have a choice x^1 such that $p \cdot x^1 = w^1$.

(a) Determine the region of permissible choices x^1 if the choices x^0 and x^1 are consistent with maximization of preferences.

(b) Determine the region of permissible choices x^1 if the choices x^0 and x^1 are consistent with maximization of preferences that are quasilinear with respect to the *first* good.

(c) Determine the region of permissible choices x^1 if the choices x^0 and x^1 are consistent with maximization of preferences that are quasilinear with respect to the *second* good.

(d) Determine the region of permissible choices x^1 if the choices x^0 and x^1 are consistent with maximization of preferences for which both goods are normal.

(e) Determine the region of permissible choices x^1 if the choices x^0 and x^1 are consistent with maximization of homothetic preferences.

[Hint: The ideal way to answer this exercise relies on (good) pictures as much as possible.]

3.D.8^A Show that for all (p, w) , $w \partial v(p, w) / \partial w = -p \cdot \nabla_p v(p, w)$.

3.E.1^A In text.

3.E.2^A In text.

3.E.3^B Prove that a solution to the EMP exists if $p \gg 0$ and there is some $x \in \mathbb{R}_+^L$ satisfying $u(x) \geq u$.

3.E.4^B Show that if the consumer's preferences $\succsim$ are convex, then $h(p, u)$ is a convex set. Also show that if $u(x)$ is strictly convex, then $h(p, u)$ is single-valued.

3.E.5^B Show that if $u(\cdot)$ is homogeneous of degree one, then $h(p, u)$ and $e(p, u)$ are homogeneous of degree one in u [i.e., they can be written as $h(p, u) = \tilde{h}(p)u$ and $e(p, u) = \tilde{e}(p)u$].

3.E.6^B Consider the constant elasticity of substitution utility function studied in Exercises 3.C.6 and 3.D.5 with $\alpha_1 = \alpha_2 = 1$. Derive its Hicksian demand function and expenditure function. Verify the properties of Propositions 3.E.2 and 3.E.3.

3.E.7^B Show that if $\succsim$ is quasilinear with respect to good 1, the Hicksian demand functions for goods 2, ..., L do not depend on u . What is the form of the expenditure function in this case?

3.E.8^A For the Cobb–Douglas utility function, verify that the relationships in (3.E.1) and (3.E.4) hold. Note that the expenditure function can be derived by simply inverting the indirect utility function, and vice versa.

3.E.9^B Use the relations in (3.E.1) to show that the properties of the indirect utility function identified in Proposition 3.D.3 imply Proposition 3.E.2. Likewise, use the relations in (3.E.1) to prove that Proposition 3.E.2 implies Proposition 3.D.3.

3.E.10^B Use the relations in (3.E.1) and (3.E.4) and the properties of the indirect utility and expenditure functions to show that Proposition 3.D.2 implies Proposition 3.E.4. Then use these facts to prove that Proposition 3.E.3 implies Proposition 3.D.2.

3.F.1^B Prove formally that a closed, convex set $K \subset \mathbb{R}^L$ equals the intersection of the half-spaces that contain it (use the separating hyperplane theorem).

3.F.2^A Show by means of a graphic example that the separating hyperplane theorem does not hold for nonconvex sets. Then argue that if K is closed and not convex, there is always some $x \notin K$ that cannot be separated from K .

3.G.1^B Prove that Proposition 3.G.1 is implied by Roy's identity (Proposition 3.G.4).

3.G.2^B Verify for the case of a Cobb–Douglas utility function that all of the propositions in Section 3.G hold.

3.G.3^B Consider the (linear expenditure system) utility function given in Exercise 3.D.6.

(a) Derive the Hicksian demand and expenditure functions. Check the properties listed in Propositions 3.E.2 and 3.E.3.

(b) Show that the derivatives of the expenditure function are the Hicksian demand function you derived in (a).

(c) Verify that the Slutsky equation holds.

(d) Verify that the own-substitution terms are negative and that compensated cross-price effects are symmetric.

(e) Show that $S(p, w)$ is negative semidefinite and has rank 2.

3.G.4^B A utility function $u(x)$ is *additively separable* if it has the form $u(x) = \sum_{\ell} u_{\ell}(x_{\ell})$.

(a) Show that additive separability is a cardinal property that is preserved only under linear transformations of the utility function.

(b) Show that the induced ordering on any group of commodities is independent of whatever fixed values we attach to the remaining ones. It turns out that this ordinal property is not only necessary but also sufficient for the existence of an additive separable representation. [You should *not* attempt a proof. This is very hard. See Debreu (1960)].

(c) Show that the Walrasian and Hicksian demand functions generated by an additively separable utility function admit no inferior goods if the functions $u_{\ell}(\cdot)$ are strictly concave. (You can assume differentiability and interiority to answer this question.)

(d) (Harder) Suppose that all $u_{\ell}(\cdot)$ are identical and twice differentiable. Let $\hat{u}(\cdot) = u_{\ell}(\cdot)$. Show that if $-[\hat{u}''(t)/\hat{u}'(t)] < 1$ for all t , then the Walrasian demand $x(p, w)$ has the so-called *gross substitute property*, i.e., $\partial x_{\ell}(p, w)/\partial p_k > 0$ for all ℓ and $k \neq \ell$.

3.G.5^C (*Hicksian composite commodities*.) Suppose there are two groups of desirable commodities, x and y , with corresponding prices p and q . The consumer's utility function is $u(x, y)$, and her wealth is $w > 0$. Suppose that prices for goods y always vary in proportion to one another, so that we can write $q = \alpha q_0$. For any number $z \geq 0$, define the function

$$\begin{aligned} \tilde{u}(x, z) = \text{Max}_y \quad & u(x, y) \\ \text{s.t. } & q_0 \cdot y \leq z. \end{aligned}$$

(a) Show that if we imagine that the goods in the economy are x and a single composite commodity z , that $\tilde{u}(x, z)$ is the consumer's utility function, and that α is the price of the composite commodity, then the solution to $\text{Max}_{x,z} \tilde{u}(x, z)$ s.t. $p \cdot x + \alpha z \leq w$ will give the consumer's actual levels of x and $z = q_0 \cdot y$.

(b) Show that properties of Walrasian demand functions identified in Propositions 3.D.2 and 3.G.4 hold for $x(p, \alpha, w)$ and $z(p, \alpha, w)$.

(c) Show that the properties in Propositions 3.E.3, and 3.G.1 to 3.G.3 hold for the Hicksian demand functions derived using $\tilde{u}(x, z)$.

3.G.6^B (F. M. Fisher) A consumer in a three-good economy (goods denoted x_1, x_2 , and x_3 ; prices denoted p_1, p_2, p_3) with wealth level $w > 0$ has demand functions for commodities 1 and 2 given by

$$x_1 = 100 - 5 \frac{p_1}{p_3} + \beta \frac{p_2}{p_3} + \delta \frac{w}{p_3}$$

$$x_2 = \alpha + \beta \frac{p_1}{p_3} + \gamma \frac{p_2}{p_3} + \delta \frac{w}{p_3}$$

where Greek letters are nonzero constants.

(a) Indicate how to calculate the demand for good 3 (but do not actually do it).

(b) Are the demand functions for x_1 and x_2 appropriately homogeneous?

(c) Calculate the restrictions on the numerical values of α, β, γ and δ implied by utility maximization.

(d) Given your results in part (c), for a fixed level of x_3 draw the consumer's indifference curve in the x_1, x_2 plane.

(e) What does your answer to (d) imply about the form of the consumer's utility function $u(x_1, x_2, x_3)$?

3.G.7^A A striking duality is obtained by using the concept of *indirect demand function*. Fix w at some level, say $w = 1$; from now on, we write $x(p, 1) = x(p)$, $v(p, 1) = v(p)$. The *indirect demand function* $g(x)$ is the inverse of $x(p)$; that is, it is the rule that assigns to every commodity bundle $x \gg 0$ the price vector $g(x)$ such that $x = x(g(x), 1)$. Show that

$$g(x) = \frac{1}{x \cdot \nabla u(x)} \nabla u(x).$$

Deduce from Proposition 3.G.4 that

$$x(p) = \frac{1}{p \cdot \nabla v(p)} \nabla v(p).$$

Note that this is a completely symmetric expression. Thus, direct (Walrasian) demand is the normalized derivative of indirect utility, and indirect demand is the normalized derivative of direct utility.

3.G.8^B The indirect utility function $v(p, w)$ is logarithmically homogeneous if $v(p, \alpha w) = v(p, w) + \ln \alpha$ for $\alpha > 0$ [in other words, $v(p, w) = \ln(v^*(p, w))$, where $v^*(p, w)$ is homogeneous of degree one]. Show that if $v(\cdot, \cdot)$ is logarithmically homogeneous, then $x(p, 1) = -\nabla_p v(p, 1)$.

3.G.9^C Compute the Slutsky matrix from the indirect utility function.

3.G.10^B For a function of the Gorman form $v(p, w) = a(p) + b(p)w$, which properties will the functions $a(\cdot)$ and $b(\cdot)$ have to satisfy for $v(p, w)$ to qualify as an indirect utility function?

3.G.11^B Verify that an indirect utility function in Gorman form exhibits linear wealth-expansion curves.

3.G.12^B What restrictions on the Gorman form correspond to the cases of homothetic and quasilinear preferences?

3.G.13^C Suppose that the indirect utility function $v(p, w)$ is a polynomial of degree n on w (with coefficients that may depend on p). Show that any individual wealth-expansion path is contained in a linear subspace of at most dimension $n + 1$. Interpret.

3.G.14^A The matrix below records the (Walrasian) demand substitution effects for a consumer endowed with rational preferences and consuming three goods at the prices $p_1 = 1$, $p_2 = 2$, and $p_3 = 6$:

$$\begin{bmatrix} -10 & ? & ? \\ ? & -4 & ? \\ 3 & ? & ? \end{bmatrix}.$$

Supply the missing numbers. Does the resulting matrix possess all the properties of a substitution matrix?

✓ **3.G.15^B** Consider the utility function

$$u = 2x_1^{1/2} + 4x_2^{1/2}.$$

(a) Find the demand functions for goods 1 and 2 as they depend on prices and wealth.

(b) Find the compensated demand function $h(\cdot)$.

(c) Find the expenditure function, and verify that $h(p, u) = \nabla_p e(p, u)$.

(d) Find the indirect utility function, and verify Roy's identity.

✓ **3.G.16^C** Consider the expenditure function

$$e(p, u) = \exp \left\{ \sum_{\ell} \alpha_{\ell} \log p_{\ell} + \left(\prod_{\ell} p_{\ell}^{\beta_{\ell}} \right) u \right\}.$$

(a) What restrictions on $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ are necessary for this to be derivable from utility maximization?

(b) Find the indirect utility that corresponds to it.

(c) Verify Roy's identity and the Slutsky equation.

3.G.17^B [From Hausman (1981)] Suppose $L = 2$. Consider a "local" indirect utility function defined in some neighborhood of price-wealth pair $(\bar{p}, \bar{w})$ by

$$v(p, w) = -\exp(-bp_1/p_2) \left[\frac{w}{p_2} + \frac{1}{b} \left(a \frac{p_1}{p_2} + \frac{a}{b} + c \right) \right].$$

(a) Verify that the local demand function for the first good is

$$x_1(p, w) = a \frac{p_1}{p_2} + b \frac{w}{p_2} + c.$$

(b) Verify that the local expenditure function is

$$e(p, u) = -p_2 u \exp(bp_1/p_2) - \frac{1}{b} \left(ap_1 + \frac{a}{b} p_2 + cp_2 \right).$$

(c) Verify that the local Hicksian demand function for the first commodity is

$$h_1(p, u) = -ub \exp(bp_1/p_2) - \frac{a}{b}.$$

3.G.18^C Show that every good is related to every other good by a chain of (weak) substitutes; that is, for any goods ℓ and k , either $\partial h_\ell(p, u)/\partial p_k \geq 0$, or there exists a good r such that $\partial h_\ell(p, u)/\partial p_r \geq 0$ and $\partial h_r(p, u)/\partial p_k \geq 0$, or there is $\dots$, and so on. [Hint: Argue first the case of two commodities. Use next the insights on composite commodities gained in Exercise 3.G.5 to handle the case of three, and then L , commodities.]

3.H.1^C Show that if $e(p, u)$ is continuous, increasing in u , homogeneous of degree one, nondecreasing, and concave in p , then the utility function $u(x) = \text{Sup} \{u: x \in V_u\}$, where $V_u = \{y: p \cdot y \geq e(p, u) \text{ for all } p \gg 0\}$, defined for $x \gg 0$, satisfies $e(p, u) = \text{Min} \{p \cdot x: u(x) \geq u\}$ for any $p \gg 0$.

3.H.2^B Use Proposition 3.F.1 to argue that if $e(p, u)$ is differentiable in p , then there are no (strongly monotone) nonconvex preferences generating $e(\cdot)$.

3.H.3^A How would you recover $v(p, w)$ from $e(p, u)$?

3.H.4^B Suppose that we are given as primitive, not the Walrasian demand but the indirect demand function $g(x)$ introduced in Exercise 3.G.7. How would you go about recovering $\succsim$? Restrict yourself to the case $L = 2$.

3.H.5^B Suppose you know the indirect utility function. How would you recover from it the expenditure function and the direct utility function?

3.H.6^B Suppose that you observe the Walrasian demand functions $x_\ell(p, w) = \alpha_\ell w/p_\ell$ for all $\ell = 1, \dots, L$ with $\sum_\ell \alpha_\ell = 1$. Derive the expenditure function of this demand system. What is the consumer's utility function?

3.H.7^B Answer the following questions with reference to the demand function in Exercise 2.F.17.

(a) Let the utility associated with consumption bundle $x = (1, 1, \dots, 1)$ be 1. What is the expenditure function $e(p, 1)$ associated with utility level $u = 1$? [Hint: Use the answer to (d) in Exercise 2.F.17.]

(b) What is the upper contour set of consumption bundle $x = (1, 1, \dots, 1)$?

3.I.1^B In text.

3.I.2^B In text.

3.I.3^B Consider a price change from initial price vector p^0 to new price vector $p^1 \leq p^0$ in which only the price of good ℓ changes. Show that $CV(p^0, p^1, w) > EV(p^0, p^1, w)$ if good ℓ is inferior.

3.I.4^B Construct an example in which a comparison of $CV(p^0, p^1, w)$ and $CV(p^0, p^2, w)$ does not give the correct welfare ranking of p^1 versus p^2 .

3.I.5^B Show that if $u(x)$ is quasilinear with respect to the first good (and we fix $p_1 = 1$), then $CV(p^0, p^1, w) = EV(p^0, p^1, w)$ for any (p^0, p^1, w) .

3.I.6^A Suppose there are $i = 1, \dots, I$ consumers with utility functions $u_i(x)$ and wealth w_i . We consider a change from p^0 to p^1 . Show that if $\sum_i CV_i(p^0, p^1, w_i) > 0$ then we can find $\{w'_i\}_{i=1}^I$ such that $\sum_i w'_i \leq \sum_i w_i$ and $v_i(p^1, w'_i) \geq v_i(p^0, w_i)$ for all i . That is, it is in principle possible to compensate everybody for the change in prices.

3.1.7^B There are three commodities (i.e., $L = 3$), of which the third is a numeraire (let $p_3 = 1$). The market demand function $x(p, w)$ has

$$x_1(p, w) = a + bp_1 + cp_2$$

$$x_2(p, w) = d + ep_1 + gp_2.$$

- (a) Give the parameter restrictions implied by utility maximization.
- (b) Estimate the equivalent variation for a change of prices from $(p_1, p_2) = (1, 1)$ to $(\bar{p}_1, \bar{p}_2) = (2, 2)$. Verify that without appropriate symmetry, there is no path independence. Assume symmetry for the rest of the exercise.
- (c) Let EV_1 , EV_2 , and EV be the equivalent variations for a change of prices from $(p_1, p_2) = (1, 1)$ to, respectively, $(2, 1)$, $(1, 2)$, and $(2, 2)$. Compare EV with $EV_1 + EV_2$ as a function of the parameters of the problem. Interpret.
- (d) Suppose that the price increases in (c) are due to taxes. Denote the deadweight losses for each of the three experiments by DW_1 , DW_2 , and DW . Compare DW with $DW_1 + DW_2$ as a function of the parameters of the problem.
- (e) Suppose the initial tax situation has prices $(p_1, p_2) = (1, 1)$. The government wants to raise a fixed (small) amount of revenue R through commodity taxes. Call t_1 and t_2 the tax rates for the two commodities. Determine the optimal tax rates as a function of the parameters of demand if the optimality criterion is the minimization of deadweight loss.

3.1.8^B Suppose we are in a three-commodity market (i.e. $L = 3$). Letting $p_3 = 1$, the demand functions for goods 1 and 2 are

$$x_1(p, w) = a_1 + b_1p_1 + c_1p_2 + d_1p_1p_2$$

$$x_2(p, w) = a_2 + b_2p_1 + c_2p_2 + d_2p_1p_2.$$

- (a) Note that the demand for goods 1 and 2 does not depend on wealth. Write down the most general class of utility functions whose demand has this property.
- (b) Argue that if the demand functions in (a) are generated from utility maximization, then the values of the parameters cannot be arbitrary. Write down as exhaustive a list as you can of the restrictions implied by utility maximization. Justify your answer.
- (c) Suppose that the conditions in (b) hold. The initial price situation is $p = (p_1, p_2)$, and we consider a change to $p' = (p'_1, p'_2)$. Derive a measure of welfare change generated in going from p to p' .
- (d) Let the values of the parameters be $a_1 = a_2 = 3/2$, $b_1 = c_2 = 1$, $c_1 = b_2 = 1/2$, and $d_1 = d_2 = 0$. Suppose the initial price situation is $p = (1, 1)$. Compute the equivalent variation for a move to p' for each of the following three cases: (i) $p' = (2, 1)$, (ii) $p' = (1, 2)$, and (iii) $p' = (2, 2)$. Denote the respective answers by EV_1 , EV_2 , EV_3 . Under which condition will you have $EV_3 = EV_1 + EV_2$? Discuss.

3.1.9^B In a one-consumer economy, the government is considering putting a tax of t per unit on good ℓ and rebating the proceeds to the consumer (who nonetheless does not consider the effect of her purchases on the size of the rebate). Suppose that $s_{\ell\ell}(p, w) < 0$ for all (p, w) . Show that the optimal tax (in the sense of maximizing the consumer's utility) is zero.

3.1.10^B Construct an example in which the area variation measure approach incorrectly ranks p^0 and p^1 . [Hint: Let the change from p^0 to p^1 involve a change in the price of more than one good.]

3.1.11^B Suppose that we know not only p^0 , p^1 , and x^0 but also $x^1 = x(p^1, w)$. Show that if $(p^1 - p^0) \cdot x^1 > 0$, then the consumer must be worse off at price-wealth situation (p^1, w) than at (p^0, w) . Interpret this test as a first-order approximation to the expenditure function at p^1 .

Also show that an alternative way to write this test is $p^0 \cdot (x^1 - x^0) < 0$, and depict the test for the case where $L = 2$ in (x_1, x_2) space. [Hint: Locate the point x^0 on the set $\{x \in \mathbb{R}_+^L : u(x) = u^0\}$.]

3.I.12^B Extend the compensating and equivalent variation measures of welfare change to the case of changes in both prices and wealth, so that we change from (p^0, w^0) to (p^1, w^1) . Also extend the “partial information” test developed in Section 3.I to this case.

3.J.1^C Show that when $L = 2$, $x(p, w)$ satisfies the strong axiom if and only if it satisfies the weak axiom.

3.AA.1^B Suppose that the consumption set is $X = \{x \in \mathbb{R}_+^2 : x_1 + x_2 \geq 1\}$ and the utility function is $u(x) = x_2$. Represent graphically, and show (a) that the locally cheaper consumption test fails at $(p, w) = (1, 1, 1)$ and (b) that market demand is not continuous at this point. Interpret economically.

3.AA.2^C Under the conditions of Proposition 3.AA.1, show that $h(p, u)$ is upper hemicontinuous and that $e(p, u)$ is continuous (even if we replace minimum by infimum and allow $p \geq 0$). Also, assuming that $h(p, u)$ is a function, give conditions for its differentiability.

Aggregate Demand

4.A Introduction

For most questions in economics, the aggregate behavior of consumers is more important than the behavior of any single consumer. In this chapter, we investigate the extent to which the theory presented in Chapters 1 to 3 can be applied to *aggregate demand*, a suitably defined sum of the demands arising from all the economy's consumers. There are, in fact, a number of different properties of individual demand that we might hope would also hold in the aggregate. Which ones we are interested in at any given moment depend on the particular application at hand.

In this chapter, we ask three questions about aggregate demand:

- (i) Individual demand can be expressed as a function of prices and the individual's wealth level. *When can aggregate demand be expressed as a function of prices and aggregate wealth?*
- (ii) Individual demand derived from rational preferences necessarily satisfies the weak axiom of revealed preference. *When does aggregate demand satisfy the weak axiom?* More generally, when can we apply in the aggregate the demand theory developed in Chapter 2 (especially Section 2.F)?
- (iii) Individual demand has welfare significance; from it, we can derive measures of welfare change for the consumer, as discussed in Section 3.I. *When does aggregate demand have welfare significance?* In particular, when do the welfare measures discussed in Section 3.I have meaning when they are computed from the aggregate demand function?

These three questions could, with a grain of salt, be called the *aggregation theories of*, respectively, *the econometrician*, *the positive theorist*, and *the welfare theorist*.

The econometrician is interested in the degree to which he can impose a simple structure on aggregate demand functions in estimation procedures. One aspect of these concerns, which we address here, is the extent to which aggregate demand can be accurately modeled as a function of only *aggregate* variables, such as aggregate (or, equivalently, average) consumer wealth. This question is important because the econometrician's data may be available only in an aggregate form.

The positive (behavioral) theorist, on the other hand, is interested in the degree

to which the positive restrictions of individual demand theory apply in the aggregate. This can be significant for deriving predictions from models of market equilibrium in which aggregate demand plays a central role.¹

The welfare theorist is interested in the normative implications of aggregate demand. He wants to use the measures of welfare change derived in Section 3.I to evaluate the welfare significance of changes in the economic environment. Ideally, he would like to treat aggregate demand as if it were generated by a “representative consumer” and use the changes in this fictional individual’s welfare as a measure of aggregate welfare.

Although the conditions we identify as important for each of these aggregation questions are closely related, the questions being asked in the three cases are conceptually quite distinct. Overall, we shall see that, in all three cases, very strong restrictions will need to hold for the desired aggregation properties to obtain. We discuss these three questions, in turn, in Sections 4.B to 4.D.

Finally, Appendix A discusses the regularizing (i.e., “smoothing”) effects arising from aggregation over a large number of consumers.

4.B Aggregate Demand and Aggregate Wealth

Suppose that there are I consumers with rational preference relations $\succsim_i$ and corresponding Walrasian demand functions $x_i(p, w_i)$. In general, given prices $p \in \mathbb{R}^L$ and wealth levels $(w_1, \dots, w_I)$ for the I consumers, aggregate demand can be written as

$$x(p, w_1, \dots, w_I) = \sum_{i=1}^I x_i(p, w_i).$$

Thus, aggregate demand depends not only on prices but also on the specific wealth levels of the various consumers. In this section, we ask when we are justified in writing aggregate demand in the simpler form $x(p, \sum_i w_i)$, where aggregate demand depends only on aggregate wealth $\sum_i w_i$.

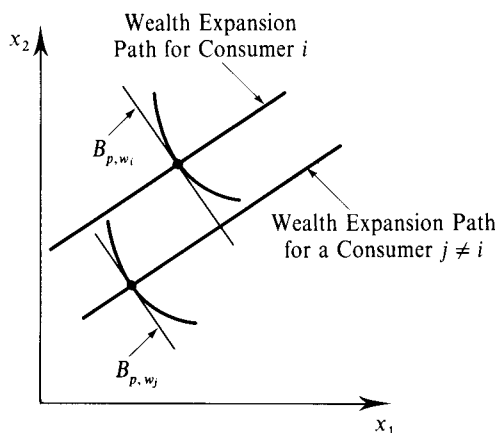
For this property to hold in all generality, aggregate demand must be identical for any two distributions of the same total amount of wealth across consumers. That is, for any $(w_1, \dots, w_I)$ and $(w'_1, \dots, w'_I)$ such that $\sum_i w_i = \sum_i w'_i$, we must have $\sum_i x_i(p, w_i) = \sum_i x_i(p, w'_i)$.

To examine when this condition is satisfied, consider, starting from some initial distribution $(w_1, \dots, w_I)$, a differential change in wealth $(dw_1, \dots, dw_I) \in \mathbb{R}^I$ satisfying $\sum_i dw_i = 0$. If aggregate demand can be written as a function of aggregate wealth, then assuming differentiability of the demand functions, we must have

$$\sum_i \frac{\partial x_{\ell i}(p, w_i)}{\partial w_i} dw_i = 0 \quad \text{for every } \ell.$$

This can be true for all redistributions $(dw_1, \dots, dw_I)$ satisfying $\sum_i dw_i = 0$ and from any initial wealth distribution $(w_1, \dots, w_I)$ if and only if the coefficients of the different

1. The econometrician may also be interested in these questions because a priori restrictions on the properties of aggregate demand can be incorporated into his estimation procedures.

**Figure 4.B.1**

Invariance of aggregate demand to redistribution of wealth implies wealth expansion paths that are straight and parallel across consumers.

dw_i are equal; that is,

$$\frac{\partial x_{\ell i}(p, w_i)}{\partial w_i} = \frac{\partial x_{\ell j}(p, w_j)}{\partial w_j} \quad (4.B.1)$$

for every ℓ , any two individuals i and j , and all $(w_1, \dots, w_I)$.²

In short, for any fixed price vector p , and any commodity ℓ , the wealth effect at p must be the same whatever consumer we look at and whatever his level of wealth.³ It is indeed fairly intuitive that in this case, the individual demand changes arising from any wealth redistribution across consumers will cancel out. Geometrically, the condition is equivalent to the statement that all consumers' wealth expansion paths are parallel, straight lines. Figure 4.B.1 depicts parallel, straight wealth expansion paths.

One special case in which this property holds arises when all consumers have identical preferences that are homothetic. Another is when all consumers have preferences that are quasilinear with respect to the same good. Both cases are examples of a more general result shown in Proposition 4.B.1.

Proposition 4.B.1: A necessary and sufficient condition for the set of consumers to exhibit parallel, straight wealth expansion paths at any price vector p is that preferences admit indirect utility functions of the Gorman form with the coefficients on w_j the same for every consumer i . That is:

$$v_i(p, w_i) = a_i(p) + b(p)w_i.$$

Proof: You are asked to establish sufficiency in Exercise 4.B.1 (this is not too difficult; use Roy's identity). Keep in mind that we are neglecting boundaries (alternatively, the significance of a result such as this is only local). You should not attempt to prove necessity. For a discussion of this result, see Deaton and Muellbauer (1980). ■

2. As usual, we are neglecting boundary constraints; hence, strictly speaking, the validity of our claims in this section is only local.

3. Note that $\partial x_{\ell i}(p, w_i)/\partial w_i = \partial x_{\ell i}(p, w'_i)/\partial w_i$ for all $w_i \neq w'_i$ because for any values of $w_j, j \neq i$, (4.B.1) must hold for the wealth distributions $(w_1, \dots, w_{i-1}, w_i, w_{i+1}, \dots, w_I)$ and $(w_1, \dots, w_{i-1}, w'_i, w_{i+1}, \dots, w_I)$. Hence, $\partial x_{\ell i}(p, w_i)/\partial w_i = \partial x_{\ell j}(p, w_j)/\partial w_j = \partial x_{\ell i}(p, w'_i)/\partial w_i$ for any $j \neq i$.

Thus, aggregate demand can be written as a function of aggregate wealth if and only if all consumers have preferences that admit indirect utility functions of the Gorman form with equal wealth coefficients $b(p)$. Needless to say, this is a very restrictive condition on preferences.⁴

Given this conclusion, we might ask whether less restrictive conditions can be obtained if we consider aggregate demand functions that depend on a wider set of aggregate variables than just the total (or, equivalently, the mean) wealth level. For example, aggregate demand might be allowed to depend on both the mean and the variance of the statistical distribution of wealth or even on the whole statistical distribution itself. Note that the latter condition is still restrictive. It implies that aggregate demand depends only on how many rich and poor there are, not on who in particular is rich or poor.

These more general forms of dependence on the distribution of wealth are indeed valid under weaker conditions than those required for aggregate demand to depend only on aggregate wealth. For a trivial example, note that aggregate demand depends only on the statistical distribution of wealth whenever all consumers possess identical but otherwise arbitrary preferences and differ only in their wealth levels. We shall not pursue this topic further here; good references are Deaton and Muellbauer (1980), Lau (1982) and Jorgenson (1990).

There is another way in which we might be able to get a more positive answer to our question. So far, the test that we have applied is whether the aggregate demand function can be written as a function of aggregate wealth for *any* distribution of wealth across consumers. The requirement that this be true for every conceivable wealth distribution is a strong one. Indeed, in many situations, individual wealth levels may be generated by some underlying process that restricts the set of individual wealth levels which can arise. If so, it may still be possible to write aggregate demand as a function of prices and aggregate wealth.

For example, when we consider general equilibrium models in Part IV, individual wealth is generated by individuals' shareholdings of firms and by their ownership of given, fixed stocks of commodities. Thus, the individual levels of real wealth are determined as a function of the prevailing price vector.

Alternatively, individual wealth levels may be determined in part by various government programs that redistribute wealth across consumers (see Section 4.D). Again, these programs may limit the set of possible wealth distributions that may arise.

To see how this can help, consider an extreme case. Suppose that individual i 's wealth level is generated by some process that can be described as a function of prices p and aggregate wealth w , $w_i(p, w)$. This was true, for example, in the general equilibrium illustration above. Similarly, the government program may base an individual's taxes (and hence his final wealth position) on his wage rate and the total (real) wealth of the society. We call a family of functions $(w_1(p, w), \dots, w_I(p, w))$ with $\sum_i w_i(p, w) = w$ for all (p, w) a *wealth distribution rule*. When individual wealth levels

4. Recall, however, that it includes some interesting and important classes of preferences. For example, if preferences are quasilinear with respect to good ℓ , then there is an indirect utility of the form $a_i(p) + w_i/p_\ell$, which, letting $b(p) = 1/p_\ell$, we can see is of the Gorman type with identical $b(p)$.

are generated by a wealth distribution rule, we can indeed *always* write aggregate demand as a function $x(p, w) = \sum_i x_i(p, w_i(p, w))$, and so aggregate demand depends only on prices and aggregate wealth.

4.C Aggregate Demand and the Weak Axiom

To what extent do the positive properties of individual demand carry over to the aggregate demand function $x(p, w_1, \dots, w_I) = \sum_i x_i(p, w_i)$? We can note immediately three properties that do: continuity, homogeneity of degree zero, and Walras' law [that is, $p \cdot x(p, w_1, \dots, w_I) = \sum_I w_i$ for all $(p, w_1, \dots, w_I)$]. In this section, we focus on the conditions under which aggregate demand also satisfies the weak axiom, arguably the most central positive property of the individual Walrasian demand function.

To study this question, we would like to operate on an aggregate demand written in the form $x(p, w)$, where w is aggregate wealth. This is the form for which we gave the definition of the weak axiom in Chapter 2. We accomplish this by supposing that there is a wealth distribution rule $(w_1(p, w), \dots, w_I(p, w))$ determining individual wealths from the price vector and total wealth. We refer to the end of Section 4.B for a discussion of wealth distribution rules.⁵ With the wealth distribution rule at our disposal, aggregate demand can automatically be written as

$$x(p, w) = \sum_i x_i(p, w_i(p, w)).$$

Formally, therefore, the aggregate demand function $x(p, w)$ depends then only on aggregate wealth and is therefore a market demand function in the sense discussed in Chapter 2.⁶ We now investigate the fulfillment of the weak axiom by $x(\cdot, \cdot)$.

In point of fact, and merely for the sake of concreteness, we shall be even more specific and focus on a particularly simple example of a distribution rule. Namely, we restrict ourselves to the case in which relative wealths of the consumers remain fixed, that is, are independent of prices. Thus, we assume that we are given wealth shares $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, so that $w_i(p, w) = \alpha_i w$ for every level $w \in \mathbb{R}$ of aggregate wealth.⁷ We have then

$$x(p, w) = \sum_i x_i(p, \alpha_i w).$$

We begin by recalling from Chapter 2 the definition of the weak axiom.

Definition 4.C.1: The aggregate demand function $x(p, w)$ satisfies the weak axiom (WA) if $p \cdot x(p', w') \leq w$ and $x(p, w) \neq x(p', w')$ imply $p' \cdot x(p, w) > w'$ for any (p, w) and (p', w') .

5. There is also a methodological advantage to assuming the presence of a wealth distribution rule. It avoids confounding different aggregation issues because the aggregation problem studied in Section 4.B (invariance of demand to redistributions) is then entirely assumed away.

6. Note that it assigns commodity bundles to price-wealth combinations, and, provided every $w_i(\cdot, \cdot)$ is continuous and homogeneous of degree one, that it is continuous, homogeneous of degree zero, and satisfies Walras's law.

7. Observe that this distribution rule amounts to leaving the wealth levels $(w_1, \dots, w_I)$ unaltered and considering only changes in the price vector p . This is because the homogeneity of degree zero of $x(p, w_1, \dots, w_I)$ implies that any proportional change in wealths can also be captured by a proportional change in prices. The description by means of shares is, however, analytically more convenient.

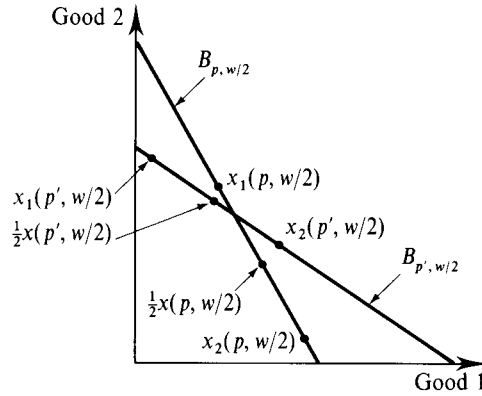


Figure 4.C.1
Failure of aggregate demand to satisfy the weak axiom.

We next provide an example illustrating that aggregate demand may not satisfy the weak axiom.

Example 4.C.1: Failure of Aggregate Demand to Satisfy the WA. Suppose that there are two commodities and two consumers. Wealth is distributed equally so that $w_1 = w_2 = w/2$, where w is aggregate wealth. Two price vectors p and p' with corresponding individual demands $x_1(p, w/2)$ and $x_2(p, w/2)$ under p , and $x_1(p', w/2)$ and $x_2(p', w/2)$ under p' , are depicted in Figure 4.C.1.

These individual demands satisfy the weak axiom, but the aggregate demands do not. Figure 4.C.1 shows the vectors $\frac{1}{2}x(p, w)$ and $\frac{1}{2}x(p', w)$, which are equal to the average of the two consumers' demands; (and so for each price vector, they must lie at the midpoint of the line segment connecting the two individuals' consumption vectors). As illustrated in the figure, we have

$$\frac{1}{2}p \cdot x(p', w) < w/2 \quad \text{and} \quad \frac{1}{2}p' \cdot x(p, w) < w/2,$$

which (multiply both sides by 2) constitutes a violation of the weak axiom at the price-wealth pairs considered. ■

The reason for the failure illustrated in Example 4.C.1 can be traced to wealth effects. Recall from Chapter 2 (Proposition 2.F.1) that $x(p, w)$ satisfies the weak axiom if and only if it satisfies the law of demand for *compensated* price changes. Precisely, if and only if for any (p, w) and any price change p' that is compensated [so that $w' = p' \cdot x(p, w)$], we have

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad (4.C.1)$$

with strict inequality if $x(p, w) \neq x(p', w')$.⁸

If the price-wealth change under consideration, say from (p, w) to (p', w') , happened to be a compensated price change for *every* consumer i —that is, if $\alpha_i w' = p' \cdot x_i(p, \alpha_i w)$ for all i —then because individual demand satisfies the weak axiom, we would know (again by Proposition 2.F.1) that for all $i = 1, \dots, I$:

$$(p' - p) \cdot [x_i(p', \alpha_i w') - x_i(p, \alpha_i w)] \leq 0, \quad (4.C.2)$$

8. Note that if $p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)$, then we must have $p' \cdot x(p, w) > w'$, in agreement with the weak axiom.

with strict inequality if $x_i(p', \alpha_i w) \neq x_i(p, \alpha_i w')$. Adding (4.C.2) over i gives us precisely (4.C.1). Thus, we conclude that aggregate demand must satisfy the WA for any price-wealth change that is compensated for every consumer.

The difficulty arises because a price-wealth change that is compensated in the aggregate, so that $w' = p' \cdot x(p, w)$, need not be compensated for each individual; we may well have $\alpha_i w' \neq p' \cdot x_i(p, \alpha_i w)$ for some or all i . If so, the individual wealth effects [which, except for the condition $p \cdot D_{w_i} x(p, \alpha_i w) = 1$, are essentially unrestricted] can play havoc with the well-behaved but possibly small individual substitution effects. The result may be that (4.C.2) fails to hold for some i , thus making possible the failure of the similar expression (4.C.1) in the aggregate.

Given that a property of individual demand as basic as the WA cannot be expected to hold generally for aggregate demand, we might wish to know whether there are any restrictions on individual preferences under which it must be satisfied. The preceding discussion suggests that it may be worth exploring the implications of assuming that the law of demand, expression (4.C.2), holds at the individual level for price changes that are left uncompensated. Suppose, indeed, that given an initial position (p, w_i) , we consider a price change p' that is not compensated, namely, we leave $w'_i = w_i$. If (4.C.2) nonetheless holds, then by addition so does (4.C.1). More formally, we begin with a definition.

Definition 4.C.2: The individual demand function $x_i(p, w_i)$ satisfies the *uncompensated law of demand (ULD)* property if

$$(p' - p) \cdot [x_i(p', w_i) - x_i(p, w_i)] \leq 0 \quad (4.C.3)$$

for any p, p' , and w_i , with strict inequality if $x_i(p', w_i) \neq x_i(p, w_i)$.

The analogous definition applies to the aggregate demand function $x(p, w)$.

In view of our discussion of the weak axiom in Section 2.F, the following differential version of the ULD property should come as no surprise (you are asked to prove it in Exercise 4.C.1):

If $x_i(p, w_i)$ satisfies the ULD property, then $D_p x_i(p, w_i)$ is negative semidefinite; that is, $dp \cdot D_p x_i(p, w_i) dp \leq 0$ for all dp .

As with the weak axiom, there is a converse to this:

If $D_p x_i(p, w_i)$ is negative definite for all p , then $x_i(p, w_i)$ satisfies the ULD property.

The analogous differential version holds for the aggregate demand function $x(p, w)$.

The great virtue of the ULD property is that, in contrast with the WA, it does, in fact, aggregate. Adding the individual condition (4.C.3) for $w_i = \alpha_i w$ gives us $(p' - p) \cdot [x(p', w) - x(p, w)] \leq 0$, with strict inequality if $x(p, w) \neq x'(p, w)$. This leads us to Proposition 4.C.1.

Proposition 4.C.1: If every consumer's Walrasian demand function $x_i(p, w_i)$ satisfies the uncompensated law of demand (ULD) property, so does the aggregate demand $x(p, w) = \sum_i x_i(p, \alpha_i w)$. As a consequence, the aggregate demand $x(p, w)$ satisfies the weak axiom.

Proof: Consider any (p, w) , (p', w) with $x(p, w) \neq x(p', w)$. We must have

$$x_i(p, \alpha_i w) \neq x_i(p', \alpha_i w)$$

for some i . Therefore, adding (4.C.3) over i , we get

$$(p' - p) \cdot [x(p, w) - x(p', w)] < 0.$$

This holds for all p, p' , and w .

To verify the WA, take any (p, w) , (p', w') with $x(p, w) \neq x(p', w')$ and $p \cdot x(p', w') \leq w$.⁹ Define $p'' = (w/w')p'$. By homogeneity of degree zero, we have $x(p'', w) = x(p', w')$. From $(p'' - p) \cdot [x(p'', w) - x(p, w)] < 0$, $p \cdot x(p'', w) \leq w$, and Walras' law, it follows that $p'' \cdot x(p, w) > w$. That is, $p' \cdot x(p, w) > w'$. ■

How restrictive is the ULD property as an axiom of individual behavior? It is clearly not implied by preference maximization (see Exercise 4.C.3). Propositions 4.C.2 and 4.C.3 provide sufficient conditions for individual demands to satisfy the ULD property.

Proposition 4.C.2: If $\succsim_i$ is homothetic, then $x_i(p, w_i)$ satisfies the uncompensated law of demand (ULD) property.

Proof: We consider the differentiable case [i.e., we assume that $x_i(p, w_i)$ is differentiable and that $\succsim_i$ is representable by a differentiable utility function]. The matrix $D_p x_i(p, w_i)$ is

$$D_p x_i(p, w_i) = S_i(p, w_i) - \frac{1}{w_i} x_i(p, w_i) x_i(p, w_i)^T,$$

where $S_i(p, w_i)$ is consumer i 's Slutsky matrix. Because $[dp \cdot x_i(p, w_i)]^2 > 0$ except when $dp \cdot x_i(p, w_i) = 0$ and $dp \cdot S_i(p, w_i) dp < 0$ except when dp is proportional to p , we can conclude that $D_p x_i(p, w_i)$ is negative definite, and so the ULD condition holds. ■

In Proposition 4.C.2, the conclusion is obtained with minimal help from the substitution effects. Those could all be arbitrarily small. The wealth effects by themselves turn out to be sufficiently well behaved. Unfortunately, the homothetic case is the only one in which this is so (see Exercise 4.C.4). More generally, for the ULD property to hold, the substitution effects (which are always well behaved) must be large enough to overcome possible “perversities” coming from the wealth effects. The intriguing result in Proposition 4.3.C [due to Mitushin and Polterovich (1978) and Milleron (1974); see Mas-Colell (1991) for an account and discussion of this result] gives a concrete expression to this relative dominance of the substitution effects.

Proposition 4.C.3: Suppose that $\succsim_i$ is defined on the consumption set $X = \mathbb{R}_+^L$ and is representable by a twice continuously differentiable concave function $u_i(\cdot)$. If

$$-\frac{x_i \cdot D^2 u_i(x_i) x_i}{x_i \cdot \nabla u_i(x_i)} < 4 \quad \text{for all } x_i,$$

then $x_i(p, w_i)$ satisfies the unrestricted law of demand (ULD) property.

9. Strictly speaking, this proof is required because although we know that the WA is equivalent to the law of demand for compensated price changes, we are now dealing with uncompensated price changes.

The proof of Proposition 4.C.3 will not be given. The courageous reader can attempt it in Exercise 4.C.5.

The condition in Proposition 4.C.3 is not an extremely stringent one. In particular, notice how amply the homothetic case fits into it (Exercise 4.C.6). So, to the question “How restrictive is the ULD property as an axiom of individual behavior?” perhaps we can answer: “restrictive, but not extremely so.”¹⁰

Note, in addition, that for the ULD property to hold for aggregate demand, it is not necessary that the ULD be satisfied at the individual level. It may arise out of aggregation itself. The example in Proposition 4.C.4, due to Hildenbrand (1983), is not very realistic, but it is nonetheless highly suggestive.

Proposition 4.C.4: Suppose that all consumers have identical preferences $\succsim$ defined on $\mathbb{R}_+^L$ [with individual demand functions denoted $\tilde{x}(p, w)$] and that individual wealth is uniformly distributed on an interval $[0, \bar{w}]$ (strictly speaking, this requires a continuum of consumers). Then the aggregate (rigorously, the average) demand function

$$x(p) = \int_0^{\bar{w}} \tilde{x}(p, w) dw$$

satisfies the unrestricted law of demand (ULD) property.

Proof: Consider the differentiable case. Take $v \neq 0$. Then

$$v \cdot Dx(p)v = \int_0^{\bar{w}} v \cdot D_p \tilde{x}(p, w)v dw.$$

Also

$$D_p \tilde{x}(p, w) = S(p, w) - D_w \tilde{x}(p, w)\tilde{x}(p, w)^T,$$

where $S(p, w)$ is the Slutsky matrix of the individual demand function $x(\cdot, \cdot)$ at (p, w) . Hence,

$$v \cdot Dx(p)v = \int_0^{\bar{w}} v \cdot S(p, w)v dw - \int_0^{\bar{w}} (v \cdot D_w \tilde{x}(p, w))(v \cdot \tilde{x}(p, w)) dw.$$

The first term of this sum is negative, unless v is proportional to p . For the second, note that

$$2(v \cdot D_w \tilde{x}(p, w))(v \cdot \tilde{x}(p, w)) = \frac{d(v \cdot \tilde{x}(p, w))^2}{dw}.$$

So

$$-\int_0^{\bar{w}} (v \cdot D_w \tilde{x}(p, w))(v \cdot \tilde{x}(p, w)) dw = -\frac{1}{2} \int_0^{\bar{w}} \frac{d(v \cdot \tilde{x}(p, w))^2}{dw} dw = -\frac{1}{2} (v \cdot \tilde{x}(p, \bar{w}))^2 \leq 0,$$

where we have used $\tilde{x}(p, 0) = 0$. Observe that the sign is negative when v is proportional to p . ■

Recall that the ULD property is additive across groups of consumers. Therefore, what we need in order to apply Proposition 4.C.4 is, not that preferences be identical, but that for every preference relation, the distribution of wealth conditional on that preference be uniform over

10. Not to misrepresent the import of this claim, we should emphasize that Proposition 4.C.1, which asserts that the ULD property is preserved under addition, holds for the price-independent distribution rules that we are considering in this section. When the distribution of real wealth may depend on prices (as it typically will in the general equilibrium applications of Part IV), then aggregate demand may violate the WA even if individual demand satisfies the ULD property (see Exercise 4.C.13). We discuss this point further in Section 17.F.

some interval that includes the level 0 (in fact, a nonincreasing density function is enough; see Exercise 4.C.7).

One lesson of Proposition 4.C.4 is that the properties of aggregate demand will depend on how preferences and wealth are distributed. We could therefore pose the problem quite generally and ask which distributional conditions on preferences and wealth will lead to satisfaction of the weak axiom by aggregate demand.¹¹

As mentioned in Section 2.F, a market demand function $x(p, w)$ can be shown to satisfy the WA if for all (p, w) , the Slutsky matrix $S(p, w)$ derived from the function $x(p, w)$ satisfies $dp \cdot S(p, w) dp < 0$ for every $dp \neq 0$ not proportional to p . We now examine when this property might hold for the aggregate demand function.

The Slutsky equation for the aggregate demand function is

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T. \quad (4.C.4)$$

Or, since $x(p, w) = \sum_i x_i(p, \alpha_i w)$,

$$S(p, w) = D_p x(p, w) + [\sum_i \alpha_i D_{w_i} x_i(p, \alpha_i w)] x(p, w)^T \quad (4.C.5)$$

Next, let $S_i(p, w_i)$ denote the individual Slutsky matrices. Adding the individual Slutsky equations gives

$$\sum_i S_i(p, \alpha_i w) = \sum_i D_p x_i(p, \alpha_i w) + \sum_i D_{w_i} x_i(p, \alpha_i w) x_i(p, \alpha_i w)^T \quad (4.C.6)$$

Since $D_p x(p, w) = \sum_i D_p x_i(p, \alpha_i w)$, we can substitute (4.C.6) into (4.C.5) to get

$$S(p, w) = \sum_i S_i(p, w_i) - \sum_i \alpha_i [D_{w_i} x_i(p, \alpha_i w) - D_w x(p, w)] \left[\frac{1}{\alpha_i} x_i(p, \alpha_i w) - x(p, w) \right]^T. \quad (4.C.7)$$

Note that because of wealth effects, the Slutsky matrix of aggregate demand is *not* the sum of the individual Slutsky matrices. The difference

$$\begin{aligned} C(p, w) &= \sum_i S_i(p, \alpha_i w) - S(p, w) \\ &= \sum_i \alpha_i [D_{w_i} x_i(p, \alpha_i w) - D_w x(p, w)] \left[\frac{1}{\alpha_i} x_i(p, \alpha_i w) - x(p, w) \right]^T \end{aligned} \quad (4.C.8)$$

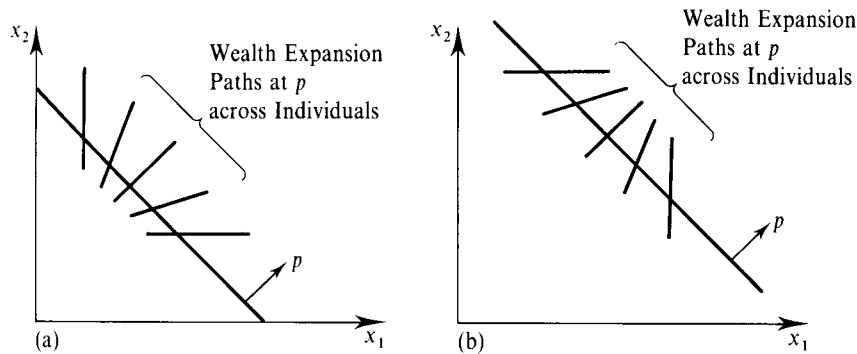
is a covariance matrix between wealth effect vectors $D_{w_i} x_i(p, \alpha_i w)$ and proportionately adjusted consumption vectors $(1/\alpha_i) x_i(p, \alpha_i w)$. The former measures how the marginal dollar is spent across commodities; the latter measures the same thing for the average dollar [e.g., $(1/\alpha_i w) x_{\ell i}(p, \alpha_i w)$ is the per-unit-of-wealth consumption of good ℓ by consumer i]. Every "observation" receives weight α_i . Note also that, as it should be, we have

$$\sum_i \alpha_i [D_{w_i} x_i(p, \alpha_i w) - D_w x(p, w)] = 0 \quad \text{and} \quad \sum_i \alpha_i [(1/\alpha_i) x_i(p, \alpha_i w) - x(p, w)] = 0.$$

For an individual Slutsky matrix $S_i(\cdot, \cdot)$ we always have $dp \cdot S_i(p, \alpha_i w) dp < 0$ for $dp \neq 0$ not proportional to p . Hence, a *sufficient* condition for the Slutsky matrix of aggregate demand to have the desired property is that $C(p, w)$ be positive semidefinite. Speaking loosely, this will be the case if, on average, there is a *positive association* across consumers between consumption (per unit of wealth) in one commodity and the wealth effect for that commodity.

Figure 4.C.2(a) depicts a case for $L = 2$ in which, assuming a uniform distribution of wealth across consumers, this association is positive: Consumers with higher-than-average

11. In the next few paragraphs, we follow Jerison (1982) and Freixas and Mas-Colell (1987).

**Figure 4.C.2**

The relation across consumers between expenditure per unit of wealth on a commodity and its wealth effect when all consumers have the same wealth.

(a) Positive relation.
(b) Negative relation.

consumption of one good spend a higher-than-average fraction of their last unit of wealth on that good. The association is negative in Figure 4.C.2(b).^{12,13}

From the preceding derivation, we can see that aggregate demand satisfies the WA in two cases of interest: (i) All the $D_{w_i}x_i(p, \alpha_i w)$ are equal (there are equal wealth effects), and (ii) all the $(1/\alpha_i)x_i(p, \alpha_i w)$ are equal (there is proportional consumption). In both cases, we have $C(p, w) = 0$, and so $dp \cdot S(p, w) dp < 0$ whenever $dp \neq 0$ is not proportional to p .

Case (i) has important implications. In particular, if every consumer has indirect utility functions of the Gorman form $v_i(p, w_i) = a_i(p) + b(p)w$, with the coefficient $b(p)$ identical across consumers, then (as we saw in Section 4.B) the wealth effects are the same for all consumers and we can therefore conclude that the WA is satisfied. We know from Section 4.B that one is led to this family of indirect utility functions by the requirement that aggregate demand be invariant to redistribution of wealth. Thus, aggregate demand satisfying the weak axiom for a fixed distribution of wealth is a less demanding property than the invariance to redistribution property considered in Section 4.B. In particular, if the second property holds, then the first also holds, but aggregate demand (for a fixed distribution of wealth) may satisfy the weak axiom even though aggregate demand may not be invariant to redistribution of wealth (e.g., individual preferences may be homothetic but not identical).

Having spent all this time investigating the weak axiom (WA), you might ask: "What about the strong axiom (SA)?" We have not focused on the Strong Axiom for three reasons.

First, the WA is a robust property, whereas the SA (which, remember, yields the symmetry of the Slutsky matrix) is not; a priori, the chances of it being satisfied by a real economy are essentially zero. For example, if we start with a group of consumers with identical preferences and wealth, then aggregate demand obviously satisfies the SA. However, if we now perturb every preference slightly and independently across consumers, the negative semidefiniteness of the Slutsky matrices (and therefore the WA) may well be preserved but the symmetry (and therefore the SA) will almost certainly not be.

12. You may want to verify that the wealth expansion paths of Example 4.C.1 must indeed look like Figure 4.C.2(b).

13. A priori, we cannot say which form is more likely. Because the demand at zero wealth is zero, it is true that for a consumer, *some* dollar must be spent among the two goods according to shares similar to the shares of the average dollar. But if the levels of wealth are not close to zero, it does not follow that this is the case for the *marginal* dollar. It may even happen that because of incipient satiation, the shares of the marginal dollar display consumption propensities that are the reverse of the ones exhibited by the average dollar. See Hildenbrand (1994) for an account of empirical research on this matter.

Second, many of the strong positive results of general equilibrium (to be reviewed in Part IV, especially Chapters 15 and 17) to which one wishes to apply the aggregation theory discussed in this chapter depend on the weak axiom, not on the strong axiom, holding in the aggregate.

Third, while one might initially think that the existence of a preference relation explaining aggregate behavior (which is what we get from the SA) would be the condition required to use aggregate demand measures (such as aggregate consumer surplus) as welfare indicators, we will see in Section 4.D that, in fact, more than this condition is required anyway.

4.D Aggregate Demand and the Existence of a Representative Consumer

The aggregation question we pose in this section is: When can we compute meaningful measures of aggregate welfare using the aggregate demand function and the welfare measurement techniques discussed in Section 3.I for individual consumers? More specifically, when can we treat the aggregate demand function as if it were generated by a fictional *representative consumer* whose preferences can be used as a measure of aggregate societal (or *social*) welfare?

We take as our starting point a distribution rule $(w_1(p, w), \dots, w_I(p, w))$ that to every level of aggregate wealth $w \in \mathbb{R}$ assigns individual wealths. We assume that $\sum_i w_i(p, w) = w$ for all (p, w) and that every $w_i(\cdot, \cdot)$ is continuous and homogeneous of degree one. As discussed in Sections 4.B and 4.C, aggregate demand then takes the form of a conventional market demand function $x(p, w) = \sum_i x_i(p, w_i(p, w))$. In particular, $x(p, w)$ is continuous, is homogeneous of degree zero, and satisfies Walras' law. It is important to keep in mind that the aggregate demand function $x(p, w)$ depends on the wealth distribution rule (except under the special conditions identified in Section 4.B).

It is useful to begin by distinguishing two senses in which we could say that there is a representative consumer. The first is a positive, or behavioral, sense.

Definition 4.D.1: A *positive representative consumer* exists if there is a rational preference relation $\succsim$ on $\mathbb{R}_+^L$ such that the aggregate demand function $x(p, w)$ is precisely the Walrasian demand function generated by this preference relation. That is, $x(p, w) \succ x$ whenever $x \neq x(p, w)$ and $p \cdot x \leq w$.

A positive representative consumer can thus be thought of as a fictional individual whose utility maximization problem when facing society's budget set $\{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$ would generate the economy's aggregate demand function.

For it to be correct to treat aggregate demand as we did individual demand functions in Section 3.I, there must be a positive representative consumer.¹⁴ However, although this is a necessary condition for the property of aggregate demand that we seek, it is not sufficient. We also need to be able to assign welfare significance to this

14. Note that if there is a positive representative consumer, then aggregate demand satisfies the positive properties sought in Section 4.C. Indeed, not only will aggregate demand satisfy the weak axiom, but it will also satisfy the strong axiom. Thus, the aggregation property we are after in this section is stronger than the one discussed in Section 4.C.

fictional individual's demand function. This will lead to the definition of a *normative* representative consumer. To do so, however, we first have to be more specific about what we mean by the term *social welfare*. We accomplish this by introducing the concept of a *social welfare function*, a function that provides a summary (social) utility index for any collection of individual utilities.

Definition 4.D.2: A (*Bergson-Samuelson*) *social welfare function* is a function $W: \mathbb{R}^I \rightarrow \mathbb{R}$ that assigns a utility value to each possible vector $(u_1, \dots, u_I) \in \mathbb{R}^I$ of utility levels for the I consumers in the economy.

The idea behind a social welfare function $W(u_1, \dots, u_I)$ is that it accurately expresses society's judgments on how individual utilities have to be compared to produce an ordering of possible social outcomes. (We do not discuss in this section the issue of where this social preference ranking comes from. Chapters 21 and 22 cover this point in much more detail.) We also assume that social welfare functions are increasing, concave, and whenever convenient, differentiable.

Let us now hypothesize that there is a process, a benevolent central authority perhaps, that, for any given prices p and aggregate wealth level w , redistributes wealth in order to maximize social welfare. That is, for any (p, w) , the wealth distribution $(w_1(p, w), \dots, w_I(p, w))$ solves

$$\begin{aligned} \text{Max}_{w_1, \dots, w_I} \quad & W(v_1(p, w_1), \dots, v_I(p, w_I)) \\ \text{s.t.} \quad & \sum_{i=1}^I w_i \leq w, \end{aligned} \quad (4.D.1)$$

where $v_i(p, w)$ is consumer i 's indirect utility function.^{15,16} The optimum value of problem (4.D.1) defines a social indirect utility function $v(p, w)$. Proposition 4.D.1 shows that this indirect utility function provides a positive representative consumer for the aggregate demand function $x(p, w) = \sum_i x_i(p, w_i(p, w))$.

Proposition 4.D.1: Suppose that for each level of prices p and aggregate wealth w , the wealth distribution $(w_1(p, w), \dots, w_I(p, w))$ solves problem (4.D.1). Then the value function $v(p, w)$ of problem (4.D.1) is an indirect utility function of a positive representative consumer for the aggregate demand function $x(p, w) = \sum_i x_i(p, w_i(p, w))$.

Proof: In Exercise 4.D.2, you are asked to establish that $v(p, w)$ does indeed have the properties of an indirect utility function. The argument for the proof then consists of using Roy's identity to derive a Walrasian demand function from $v(p, w)$, which we denote by $x_R(p, w)$, and then establishing that it actually equals $x(p, w)$.

We begin by recording the first-order conditions of problem (4.D.1) for a

15. We assume in this section that our direct utility functions $u_i(\cdot)$ are concave. This is a weak hypothesis (once quasiconcavity has been assumed) which makes sure that in all the optimization problems to be considered, the first-order conditions are sufficient for the determination of global optima. In particular, $v_i(p, \cdot)$ is then a concave function of w_i .

16. In Exercise 4.D.1, you are asked to show that if so desired, problem (4.D.1) can be equivalently formulated as one where social utility is maximized, not by distributing wealth, but by distributing bundles of goods with aggregate value at prices p not larger than w . The fact that in optimally redistributing goods, we can also restrict ourselves to redistributing wealth is, in essence, a version of the second fundamental theorem of welfare economics, which will be covered extensively in Chapter 16.

given value of (p, w) . Neglecting boundary solutions, these require that for some $\lambda \geq 0$, we have

$$\lambda = \frac{\partial W}{\partial v_1} \frac{\partial v_1}{\partial w_1} = \dots = \frac{\partial W}{\partial v_I} \frac{\partial v_I}{\partial w_I} \quad (4.D.2)$$

(For notational convenience, we have omitted the points at which the derivatives are evaluated.) Condition (4.D.2) simply says that at a socially optimal wealth distribution, the social utility of an extra unit of wealth is the same irrespective of who gets it.

By Roy's identity, we have $x_R(p, w) = -[1/(\partial v(p, w)/\partial w)] \nabla_p v(p, w)$. Since $v(p, w)$ is the value function of problem (4.D.1), we know that $\partial v/\partial w = \lambda$. (See Section M.K of the Mathematical Appendix) In addition, for any commodity ℓ , the chain rule and (4.D.2)—or, equivalently, the envelope theorem—give us

$$\frac{\partial v}{\partial p_\ell} = \sum_i \frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial p_\ell} + \lambda \sum_i \frac{\partial w_i}{\partial p_\ell} = \sum_i \frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial p_\ell},$$

where the second equality follows because $\sum_i w_i(p, w) = w$ for all (p, w) implies that $\sum_i (\partial w_i / \partial p_\ell) = 0$. Hence, in matrix notation, we have

$$\nabla_p v(p, w) = \sum_i (\partial W / \partial v_i) \nabla_p v_i(p, w_i(p, w)).$$

Finally, using Roy's identity and the first-order condition (4.D.2), we get

$$\begin{aligned} x_R(p, w) &= -\frac{1}{\lambda} \sum_i \left[\frac{\lambda}{\partial v_i / \partial w_i} \right] \nabla_p v_i(p, w_i(p, w)) \\ &= -\sum_i \left[\frac{1}{\partial v_i / \partial w_i} \right] \nabla_p v_i(p, w_i(p, w)) \\ &= \sum_i x_i(p, w_i(p, w)) = x(p, w), \end{aligned}$$

as we wanted to show. ■

Equipped with Proposition 4.D.1, we can now define a *normative representative consumer*.

Definition 4.D.3: The positive representative consumer $\succsim$ for the aggregate demand $x(p, w) = \sum_i x_i(p, w_i(p, w))$ is a *normative representative consumer* relative to the social welfare function $W(\cdot)$ if for every (p, w) , the distribution of wealth $(w_1(p, w), \dots, w_I(p, w))$ solves problems (4.D.1) and, therefore, the value function of problem (4.D.1) is an indirect utility function for $\succsim$.

If there is a normative representative consumer, the preferences of this consumer have welfare significance and the aggregate demand function $x(p, w)$ can be used to make welfare judgments by means of the techniques described in Section 3.I. In doing so, however, it should never be forgotten that a given wealth distribution rule [the one that solves (4.D.1) for the given social welfare function] is being adhered to and that the “level of wealth” should always be understood as the “optimally distributed level of wealth.” For further discussion, see Samuelson (1956) and Chipman and Moore (1979).

Example 4.D.1: Suppose that consumers all have homothetic preferences represented by utility functions homogeneous of degree one. Consider now the social welfare function $W(u_1, \dots, u_I) = \sum_i \alpha_i \ln u_i$ with $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. Then the optimal

wealth distribution function [for problem (4.D.1)] is the price-independent rule that we adopted in Section 4.C: $w_i(p, w) = \alpha_i w$. (You are asked to demonstrate this fact in Exercise 4.D.6.) Therefore, in the homothetic case, the aggregate demand $x(p, w) = \sum_i x_i(p, \alpha_i w)$ can be viewed as originating from the normative representative consumer generated by this social welfare function. ■

Example 4.D.2: Suppose that all consumers' preferences have indirect utilities of the Gorman form $v_i(p, w_i) = a_i(p) + b(p)w_i$. Note that $b(p)$ does not depend on i , and recall that this includes as a particular case the situation in which preferences are quasilinear with respect to a common numeraire. From Section 4.B, we also know that aggregate demand $x(p, w)$ is independent of the distribution of wealth.¹⁷

Consider now the *utilitarian* social welfare function $\sum_i u_i$. Then *any* wealth distribution rule $(w_1(p, w), \dots, w_I(p, w))$ solves the optimization problem (4.D.1), and the indirect utility function that this problem generates is simply $v(p, w) = \sum_i a_i(p) + b(p)w$. (You are asked to show these facts in Exercise 4.D.7.) One conclusion is, therefore, that when indirect utility functions have the Gorman form [with common $b(p)$] and the social welfare function is utilitarian, then aggregate demand can *always* be viewed as being generated by a normative representative consumer.

When consumers have Gorman-form indirect utility functions [with common $b(p)$], the theory of the normative representative consumer admits an important strengthening. In general, the preferences of the representative consumer depend on the form of the social welfare function. *But not in this case.* We now verify that if the indirect utility functions of the consumers have the Gorman form [with common $b(p)$], then the preferences of the representative consumer are independent of the particular social welfare function used.¹⁸ In fact, we show that $v(p, w) = \sum_i a_i(p) + b(p)w$ is an admissible indirect utility function for the normative representative consumer relative to *any* social welfare function $W(u_1, \dots, u_I)$.

To verify this claim, consider a particular social welfare function $W(\cdot)$, and denote the value function of problem (4.D.1), relative to $W(\cdot)$, by $v^*(p, w)$. We must show that the ordering induced by $v(\cdot)$ and $v^*(\cdot)$ is the same, that is, that for any pair (p, w) and (p', w') with $v(p, w) < v(p', w')$, we have $v^*(p, w) < v^*(p', w')$. Take the vectors of individual wealths $(w_1, \dots, w_I)$ and $(w'_1, \dots, w'_I)$ reached as optima of (4.D.1), relative to $W(\cdot)$, for (p, w) and (p', w') , respectively. Denote $u_i = a_i(p) + b(p)w_i$, $u'_i = a_i(p') + b(p')w'_i$, $u = (u_1, \dots, u_I)$, and $u' = (u'_1, \dots, u'_I)$. Then $v^*(p, w) = W(u)$ and $v^*(p', w') = W(u')$. Also $v(p, w) = \sum_i a_i(p) + b(p)w = \sum_i u_i$, and similarly, $v(p', w') = \sum_i u'_i$. Therefore, $v(p, w) < v(p', w')$ implies $\sum_i u_i < \sum_i u'_i$. We argue that $\nabla W(u') \cdot (u - u') < 0$, which, $W(\cdot)$ being concave, implies the desired result, namely $W(u) < W(u')$.¹⁹ By expression (4.D.2), at an optimum we have $(\partial W / \partial v_i)(\partial v_i / \partial w_i) = \lambda$ for all i . But in our case, $\partial v_i / \partial w_i = b(p)$ for all i . Therefore, $\partial W / \partial v_i = \partial W / \partial v_j > 0$ for any i, j . Hence, $\sum_i u_i < \sum_i u'_i$ implies $\nabla W(u') \cdot (u - u') < 0$.

The previous point can perhaps be better understood if we observe that when

17. As usual, we neglect the nonnegativity constraints on consumption.

18. But, of course, the optimal distribution rules will typically depend on the social welfare function. Only for the utilitarian social welfare function will it not matter how wealth is distributed.

19. Indeed, concavity of $W(\cdot)$ implies $W(u') + \nabla W(u') \cdot (u - u') \geq W(u)$; see Section M.C of the Mathematical Appendix.

preferences have the Gorman form [with common $b(p)$], then (p', w') is socially better than (p, w) for the utilitarian social welfare function $\sum_i u_i$ if and only if when compared with (p, w) , (p', w') passes the following *potential compensation test*: For any distribution $(w_1, \dots, w_I)$ of w , there is a distribution $(w'_1, \dots, w'_I)$ of w' such that $v_i(p', w'_i) > v_i(p, w_i)$ for all i . To verify this is straightforward. Suppose that

$$(\sum_i a_i(p') + b(p')w') - (\sum_i a_i(p) + b(p)w) = c > 0.$$

Then the wealth levels w'_i implicitly defined by $a_i(p') + b(p')w'_i = a_i(p) + b(p)w_i + c/I$ will be as desired.²⁰ Once we know that (p', w') when compared with (p, w) passes the potential compensation test, it follows merely from the definition of the optimization problem (4.D.1) that (p', w') is better than (p, w) for any normative consumer, that is, for any social welfare function that we may wish to employ (see Exercise 4.D.8).

The two properties just presented—*independence of the representative consumer's preferences from the social welfare function and the potential compensation criterion*—will be discussed further in Sections 10.F and 22.C. For the moment, we simply emphasize that they are not general properties of normative representative consumers. By choosing the distribution rules that solve (4.D.1), we can generate a normative representative consumer for any set of individual utilities and any social welfare function. For the properties just reviewed to hold, the individual preferences have been required to have the Gorman form [with common $b(p)$]. ■

It is important to stress the distinction between the concepts of a positive and a normative representative consumer. It is *not* true that whenever aggregate demand can be generated by a positive representative consumer, this representative consumer's preferences have normative content. It may even be the case that a positive representative consumer exists but that there is *no* social welfare function that leads to a normative representative consumer. We expand on this point in the next few paragraphs [see also Dow and Werlang (1988) and Jerison (1994)].

We are given a distribution rule $(w_1(p, w), \dots, w_I(p, w))$ and assume that a positive representative consumer with utility function $u(x)$ exists for the aggregate demand $x(p, w) = \sum_i x_i(p, w_i(p, w))$. In principle, using the integrability techniques presented in Section 3.H, it should be possible to determine the preferences of the representative consumer from the knowledge of $x(p, w)$. Now fix any $(\bar{p}, \bar{w})$, and let $\bar{x} = x(\bar{p}, \bar{w})$. Relative to the aggregate consumption vector $\bar{x}$, we can define an at-least-as-good-as set for the representative consumer:

$$B = \{x \in \mathbb{R}_+^L : u(x) \geq u(\bar{x})\} \subset \mathbb{R}_+^L.$$

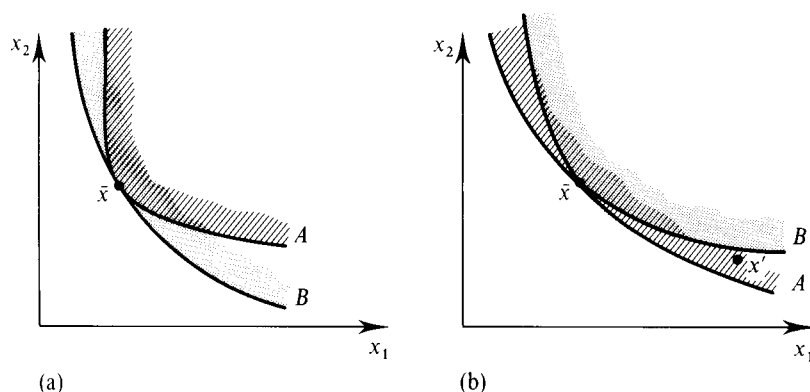
Next, let $\bar{w}_i = w_i(\bar{p}, \bar{w})$ and $\bar{x}_i = x_i(\bar{p}, \bar{w}_i)$, and consider the set

$$A = \{x = \sum_i x_i : x_i \succeq_i \bar{x}_i \text{ for all } i\} \subset \mathbb{R}_+^L.$$

In words, A is the set of aggregate consumption vectors for which there is a distribution of commodities among consumers that makes every consumer as well off as under $(\bar{x}_1, \dots, \bar{x}_I)$. The boundary of this set is sometimes called a *Scitovsky contour*. Note that both set A and set B are supported by the price vector $\bar{p}$ at $\bar{x}$ (see Figure 4.D.1).

If the given wealth distribution comes from the solution to a social welfare optimization problem of the type (4.D.1) (i.e., if the positive representative consumer is in fact a normative

20. We continue to neglect nonnegativity constraints on wealth.

**Figure 4.D.1**

Comparing the at-least-as-good-as set of the positive representative consumer with the sum of the at-least-as-good-as sets of the individual consumers.

(a) The positive representative consumer could be a normative representative consumer.

(b) The positive representative consumer cannot be a normative representative consumer.

$$A = \{x \in \sum_i x_i : u_i(x_i) \geq u_i(\bar{x}_i) \text{ for all } i\}$$

$$B = \{x \in \mathbb{R}_+^2 : u(x) \geq u(\bar{x})\}$$

representative consumer), then this places an important restriction on how sets A and B relate to each other: Every element of set A must be an element of set B . This is so because the social welfare function underlying the normative representative consumer is increasing in the utility level of every consumer (and thus any aggregate consumption bundle that could be distributed in a manner that guarantees to every consumer a level of utility as high as the levels corresponding to the optimal distribution of $\bar{x}$ must receive a social utility higher than the latter; see Exercise 4.D.4). That is, a *necessary* condition for the existence of a normative representative consumer is that $A \subset B$. A case that satisfies this necessary condition is depicted in Figure 4.D.1(a).

However, there is nothing to prevent the existence, in a particular setting, of a positive representative consumer with a utility function $u(x)$ that fails to satisfy this condition, as in Figure 4.D.1(b). To provide some further understanding of this point, Exercise 4.D.9 asks you to show that $A \subset B$ implies that $\sum_i S_i(\bar{p}, \bar{w}_i) - S(\bar{p}, \bar{w})$ is positive semidefinite, where $S(p, w)$ and $S_i(p, w_i)$ are the Slutsky matrices of aggregate and individual demand, respectively. Informally, we could say that the substitution effects of aggregate demand must be larger in absolute value than the sum of individual substitution effects (geometrically, this corresponds to the boundary of B being flatter at $\bar{x}$ than the boundary of A). This observation allows us to generate in a simple manner examples in which aggregate demand can be rationalized by preferences but, nonetheless, there is no normative representative consumer.

Suppose, for example, that the wealth distribution rule is of the form $w_i(p, w) = \alpha_i w$. Suppose also that $S(p, w)$ happens to be symmetric for all (p, w) ; if $L = 2$, this is automatically satisfied. Then, from integrability theory (see Section 3.H), we know that a sufficient condition for the existence of underlying preferences is that, for all (p, w) , we have $dp \cdot S(p, w) dp < 0$ for all $dp \neq 0$ not proportional to p (we abbreviate this as the *n.d. property*). On the other hand, as we have just seen, a necessary condition for the existence of a normative representative consumer is that $C(\bar{p}, \bar{w}) = \sum_i S_i(\bar{p}, \bar{w}_i) - S(\bar{p}, \bar{w})$ be positive semidefinite [this is the same matrix discussed in Section 4.C; see expression (4.C.8)]. Thus, if $S(p, w)$ has the n.d. property for all (p, w) but $C(\bar{p}, \bar{w})$ is not positive semidefinite [i.e., wealth effects are such that $S(\bar{p}, \bar{w})$ is “less negative” than $\sum_i S_i(\bar{p}, \bar{w}_i)$], then a positive representative consumer exists that, nonetheless, cannot be made normative for any social welfare function. (Exercise 4.D.10 provides an instance where this is indeed the case.) In any example of this nature we have moves in aggregate consumption that would pass a potential compensation test (each consumer’s welfare could be made better off by an appropriate distribution of the move) but are regarded as socially inferior under the utility function that rationalizes aggregate demand. [In Figure 4.D.1(b), this could be the move from $\bar{x}$ to x' .]

The moral of all this is clear: The existence of preferences that explain behavior is not

enough to attach to them any welfare significance. For the latter, it is also necessary that these preferences exist for the right reasons. ■

APPENDIX A: REGULARIZING EFFECTS OF AGGREGATION

This appendix is devoted to making the point that although aggregation can be deleterious to the preservation of the good properties of individual demand, it can also have helpful *regularizing* effects. By regularizing, we mean that the average (per-consumer) demand will tend to be more continuous or smooth, as a function of prices, than the individual components of the sum.

Recall that if preferences are strictly convex, individual demand functions are continuous. As we noted, aggregate demand will then be continuous as well. But average demand can be (nearly) continuous even when individual demands are not. The key requirement is one of *dispersion* of individual preferences.

Example 4.AA.1: Suppose that there are two commodities. Consumers have quasi-linear preferences with the second good as numeraire. The first good, on the other hand, is available only in integer amounts, and consumers have no wish for more than one unit of it. Thus, normalizing the utility of zero units of the first good to be zero, the preferences of consumer i are completely described by a number v_{1i} , the utility in terms of numeraire of holding one unit of the first good. It is then clear that the demand for the first good by consumer i is given by the correspondence

$$\begin{aligned} x_{1i}(p_1) &= 1 && \text{if } p_1 < v_{1i}, \\ &= \{0, 1\} && \text{if } p_1 = v_{1i}, \\ &= 0 && \text{if } p_1 > v_{1i}, \end{aligned}$$

which is depicted in Figure 4.AA.1(a). Thus, individual demand exhibits a sudden, discontinuous jump in demand from 0 to 1 as the price crosses the value $p_1 = v_{1i}$.

Suppose now that there are many consumers. In fact, consider the limit situation where there is an actual continuum of consumers. We could then say that individual preferences are *dispersed* if there is no concentrated group of consumers having any particular value of v_1 or, more precisely, if the statistical distribution function of the v_1 's, $G(v_1)$, is continuous. Then, denoting by $x_1(p_1)$ the average demand for the first good, we have $x_1(p_1) = \text{"mass of consumers with } v_1 > p_1\text{"} = 1 - G(p_1)$.

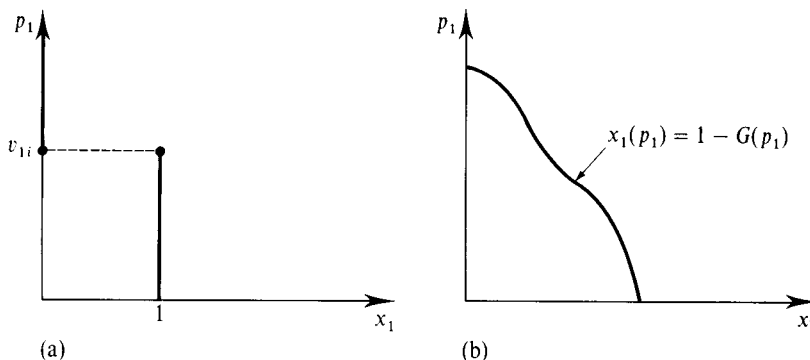


Figure 4.AA.1

The regularizing effect of aggregation.
(a) Individual demand.
(b) Aggregate demand when the distribution of the v_1 's is $G(\cdot)$.

Hence, the aggregate demand $x_1(\cdot)$, shown in Figure 4.AA.1(b), is a nice continuous function even though none of the individual demand correspondences are so. Note that with only a finite number of consumers, the distribution function $G(\cdot)$ cannot quite be a continuous function; but if the consumers are many, then it can be nearly continuous. ■

The regularizing effects of aggregation are studied again in Section 17.I. We show there that in general (i.e., without dispersedness requirements), the aggregation of numerous individual demand correspondences will generate a (nearly) *convex-valued* average demand correspondence.

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EXERCISES

4.B.1^B Prove the sufficiency part of Proposition 4.B.1. Show also that if preferences admit Gorman-form indirect utility functions with the same $b(p)$, then preferences admit expenditure functions of the form $e_i(p, u_i) = c(p)u_i + d_i(p)$.

4.B.2^B Suppose that there are I consumers and L commodities. Consumers differ only by their wealth levels w_i and by a taste parameter s_i , which we could call *family size*. Thus, denote the indirect utility function of consumer i by $v(p, w_i, s_i)$. The corresponding Walrasian demand function for consumer i is $x(p, w_i, s_i)$.

(a) Fix $(s_1, \dots, s_I)$. Show that if for any $(w_1, \dots, w_I)$ aggregate demand can be written as a function of only p and aggregate wealth $w = \sum_i w_i$ (or, equivalently, average wealth), and if every consumer's preference relationship $\succsim_i$ is homothetic, then all these preferences must be identical [and so $x(p, w_i, s_i)$ must be independent of s_i].

(b) Give a sufficient condition for aggregate demand to depend only on aggregate wealth w and $\sum_i s_i$ (or, equivalently, average wealth and average family size).

4.C.1^C Prove that if $x_i(p, w_i)$ satisfies the ULD, then $D_p x_i(p, w_i)$ is negative semidefinite [i.e., $dp \cdot D_p x_i(p, w_i) dp \leq 0$ for all dp]. Also show that if $D_p x_i(p, w_i)$ is negative definite for all p , then $x_i(p, w_i)$ satisfies the ULD (this second part is harder).

4.C.2^A Prove a version of Proposition 4.C.1 by using the (sufficient) differential versions of the ULD and the WA. (Recall from the small type part of Section 2.F that a sufficient condition for the WA is that $v \cdot S(p, w)v < 0$ whenever v is not proportional to p .)

4.C.3^A Give a graphical two-commodity example of a preference relation generating a Walrasian demand that does not satisfy the ULD property. Interpret.

4.C.4^C Show that if the preference relation $\succsim_i$ on $\mathbb{R}_+^2$ has L-shaped indifference curves and the demand function $x_i(p, w_i)$ has the ULD property, then $\succsim_i$ must be homothetic. [Hint: The L shape of indifference curves implies $S_i(p, w_i) = 0$ for all (p, w_i) ; show that if $D_{w_i} x_i(\bar{p}, \bar{w}_i) \neq (1/\bar{w}_i)x_i(\bar{p}, \bar{w}_i)$, then there is $v \in \mathbb{R}^L$ such that $v \cdot D_p x_i(\bar{p}, \bar{w}_i)v > 0$.]

4.C.5^C Prove Proposition 4.C.3. To that effect, you can fix $w = 1$. The proof is best done in terms of the indirect demand function $g_i(x) = (1/x \cdot \nabla u_i(x)) \nabla u_i(x)$ [note that $x = x_i(g_i(x), 1)$]. For an individual consumer, the ULD is self-dual; that is, it is equivalent to $(g_i(x) - g_i(y)) \cdot (x - y) < 0$ for all $x \neq y$. In turn, this property is implied by the negative definiteness of $Dg_i(x)$ for all x . Hence, concentrate on proving this last property. More specifically, let $v \neq 0$, and denote $q = \nabla u_i(x)$ and $C = D^2 u_i(x)$. You want to prove $v \cdot Dg_i(x)v < 0$. [Hint: You can first assume $q \cdot v = q \cdot x$; then differentiate $g_i(x)$, and make use of the equality $v \cdot Cv - x \cdot Cv = (v - \frac{1}{2}x) \cdot C(v - \frac{1}{2}x) - \frac{1}{4}x \cdot Cx$.]

4.C.6^A Show that if $u_i(x_i)$ is homogeneous of degree one, so that $\succsim_i$ is homothetic, then $\sigma_i(x_i) = 0$ for all x_i [$\sigma_i(x_i)$ is the quotient defined in Proposition 4.C.3].

4.C.7^B Show that Proposition 4.C.4 still holds if the distribution of wealth has a nonincreasing density function on $[0, \bar{w}]$. A more realistic distribution of wealth would be *unimodal* (i.e., an increasing and then decreasing density function with a single peak). Argue that there are unimodal distributions for which the conclusions of the proposition do not hold.

4.C.8^A Derive expression (4.C.7), the aggregate version of the Slutsky matrix.

4.C.9^A Verify that if individual preferences $\succsim_i$ are homothetic, then the matrix $C(p, w)$ defined in expression (4.C.8) is positive semidefinite.

4.C.10^C Argue that for the Hildenbrand example studied in Proposition 4.C.4, $C(p, w)$ is positive semidefinite. Conclude that aggregate demand satisfies the WA for that wealth distribution. [Note: You must first adapt the definition of $C(p, w)$ to the continuum-of-consumers situation of the example.]

4.C.11^B Suppose there are two consumers, 1 and 2, with utility functions over two goods, 1 and 2, of $u_1(x_{11}, x_{21}) = x_{11} + 4\sqrt{x_{21}}$ and $u_2(x_{12}, x_{22}) = 4\sqrt{x_{12}} + x_{22}$. The two consumers have identical wealth levels $w_1 = w_2 = w/2$.

(a) Calculate the individual demand functions and the aggregate demand function.

(b) Compute the individual Slutsky matrices $S_i(p, w/2)$ (for $i = 1, 2$) and the aggregate

Slutsky matrix $S(p, w)$. [Hint: Note that for this two-good example, only one element of each matrix must be computed to determine the entire matrix.] Show that $dp \cdot S(p, w) dp < 0$ for all $dp \neq 0$ not proportional to p . Conclude that aggregate demand satisfies the WA.

(c) Compute the matrix $C(p, w) = \sum_i S_i(p, w/2) - S(p, w)$ for prices $p_1 = p_2 = 1$. Show that it is positive semidefinite if $w > 16$ and that it is negative semidefinite if $8 < w < 16$. In fact, argue that in the latter case, $dp \cdot C(p, w) dp < 0$ for some dp [so that $C(p, w)$ is not positive semidefinite]. Conclude that $C(p, w)$ positive semidefinite is not necessary for the WA to be satisfied.

(d) For each of the two cases $w > 16$ and $8 < w < 16$, draw a picture in the (x_1, x_2) plane depicting each consumer's consumption bundle and his wealth expansion path for the prices $p_1 = p_2 = 1$. Compare your picture with Figure 4.C.2.

4.C.12^B The results presented in Sections 4.B and 4.C indicate that if for any $(w_1, \dots, w_I)$ aggregate demand can be written as a function of only aggregate wealth [i.e., as $x(p, \sum_i w_i)$], then aggregate demand must satisfy the WA. The *distribution function* $F: [0, \infty) \rightarrow [0, 1]$ of $(w_1, \dots, w_I)$ is defined as $F(w) = (1/I)(\text{number of } i\text{'s with } w_i \leq w)$ for any w . Suppose now that for any $(w_1, \dots, w_I)$, aggregate demand can be written as a function of the corresponding aggregate *distribution* $F(\cdot)$ of wealth. Show that aggregate demand does not necessarily satisfy the WA. [Hint: It suffices to give a two-commodity, two-consumer example where preferences are identical, wealths are $w_1 = 1$ and $w_2 = 3$, and the WA fails. Try to construct the example graphically. It is a matter of making sure that four suitably positioned indifference curves can be fitted together without crossing.]

4.C.13^C Consider a two-good environment with two consumers. Let the wealth distribution rule be $w_1(p, w) = wp_1/(p_1 + p_2)$, $w_2(p, w) = wp_2/(p_1 + p_2)$. Exhibit an example in which the two consumers have homothetic preferences but, nonetheless, the aggregate demand fails to satisfy the weak axiom. A good picture will suffice. Why does not Proposition 4.C.1 apply?

4.D.1^B In this question we are concerned with a normative representative consumer. Denote by $v(p, w)$ the optimal value of problem (4.D.1), and by $(w_1(p, w), \dots, w_I(p, w))$ the corresponding optimal wealth distribution rules. Verify that $v(p, w)$ is also the optimal value of

$$\begin{aligned} \text{Max}_{x_1, \dots, x_I} \quad & W(u_1(x_1), \dots, u_I(x_I)) \\ \text{s.t.} \quad & p \cdot (\sum_i x_i) \leq w \end{aligned}$$

and that $[x_1(p, w_1(p, w)), \dots, x_I(p, w_I(p, w))]$ is a solution to this latter problem. Note the implication that to maximize social welfare given prices p and wealth w , the planner need not control consumption directly, but rather need only distribute wealth optimally and allow consumers to make consumption decisions independently given prices p .

4.D.2^B Verify that $v(p, w)$, defined as the optimal value of problem (4.D.1), has the properties of an indirect utility function (i.e., that it is homogeneous of degree zero, increasing in w and decreasing in p , and quasiconvex).

4.D.3^B It is good to train one's hand in the use of inequalities and the Kuhn–Tucker conditions. Prove Proposition 4.D.1 again, this time allowing for corner solutions.

4.D.4^C Suppose that there is a normative representative consumer with wealth distribution rule $(w_1(p, w), \dots, w_I(p, w))$. For any $x \in \mathbb{R}_+^I$, define

$$\begin{aligned} u(x) = \text{Max}_{(x_1, \dots, x_I)} \quad & W(u_1(x_1), \dots, u_I(x_I)) \\ \text{s.t.} \quad & \sum_i x_i \leq x. \end{aligned}$$

(a) Give conditions implying that $u(\cdot)$ has the properties of a utility function; that is, it is monotone, continuous, and quasiconcave (and even concave).

(b) Show that for any (p, w) , the Walrasian demand generated from the problem $\text{Max}_x u(x)$ s.t. $p \cdot x \leq w$ is equal to the aggregate demand function.

4.D.5^A Suppose that there are I consumers and that consumer i 's utility function is $u_i(x_i)$, with demand function $x_i(p, w_i)$. Consumer i 's wealth w_i is generated according to a wealth distribution rule $w_i = \alpha_i w$, where $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$. Provide an example (i.e., a set of utility functions) in which this economy does *not* admit a positive representative consumer.

4.D.6^B Establish the claims made in Example 4.D.1.

4.D.7^B Establish the claims made in the second paragraph of Example 4.D.2.

4.D.8^A Say that (p', w') passes the *potential compensation test* over (p, w) if for any distribution $(w_1, \dots, w_I)$ of w there is a distribution $(w'_1, \dots, w'_I)$ of w' such that $v_i(p', w'_i) > v_i(p, w_i)$ for all i . Show that if (p', w') passes the potential compensation test over (p, w) , any normative representative consumer must prefer (p', w') over (p, w) .

4.D.9^B Show that $A \subset B$ (notation as in Section 4.D) implies that $S(\bar{p}, \bar{w}) - \sum_i S_i(\bar{p}, \bar{w}_i)$ is negative semidefinite. [Hint: Consider $g(p) = e(p, u(\bar{x})) - \sum_i e_i(p, u_i(\bar{x}_i))$, where $e(\cdot)$ is the expenditure function for $u(\cdot)$ and $e_i(\cdot)$ is the expenditure function for $u_i(\cdot)$. Note that $A = \sum_i \{x_i : u_i(x_i) \geq u_i(\bar{x}_i)\}$ implies that $\sum_i e_i(p, u_i(\bar{x}_i))$ is the optimal value of the problem $\text{Min}_{x \in A} p \cdot x$. From this and $A \subset B$, you get $g(p) \leq 0$ for all p and $g(\bar{p}) = 0$. Therefore, $D^2 g(\bar{p})$ is negative semidefinite. Show then that $D^2 g(\bar{p}) = S(\bar{p}, \bar{w}) - \sum_i S_i(\bar{p}, \bar{w}_i)$.]

4.D.10^A Argue that in the example considered in Exercise 4.C.11, there is a positive representative consumer rationalizing aggregate demand but that there cannot be a normative representative consumer.

4.D.11^C Argue that for $L > 2$, the Hildenbrand case of Proposition 4.C.4 need not admit a positive representative consumer. [Hint: Argue that the Slutsky matrix may fail to be symmetric.]

Production

5.A Introduction

In this chapter, we move to the supply side of the economy, studying the process by which the goods and services consumed by individuals are produced. We view the supply side as composed of a number of productive units, or, as we shall call them, “firms.” Firms may be corporations or other legally recognized businesses. But they must also represent the productive possibilities of individuals or households. Moreover, the set of all firms may include some potential productive units that are never actually organized. Thus, the theory will be able to accommodate both active production processes and potential but inactive ones.

Many aspects enter a full description of a firm: Who owns it? Who manages it? How is it managed? How is it organized? What can it do? Of all these questions, we concentrate on the last one. Our justification is not that the other questions are not interesting (indeed, they are), but that we want to arrive as quickly as possible at a minimal conceptual apparatus that allows us to analyze market behavior. Thus, our model of production possibilities is going to be very parsimonious: The firm is viewed merely as a “black box”, able to transform inputs into outputs.

In Section 5.B, we begin by introducing the firm’s *production set*, a set that represents the production *activities*, or *production plans*, that are technologically feasible for the firm. We then enumerate and discuss some commonly assumed properties of production sets, introducing concepts such as *returns to scale*, *free disposal*, and *free entry*.

After studying the firm’s technological possibilities in Section 5.B, we introduce its objective, the goal of *profit maximization*, in Section 5.C. We then formulate and study the firm’s profit maximization problem and two associated objects, the firm’s *profit function* and its *supply correspondence*. These are, respectively, the value function and the optimizing vectors of the firm’s profit maximization problem. Related to the firm’s goal of profit maximization is the task of achieving cost-minimizing production. We also study the firm’s cost minimization problem and two objects associated with it: The firm’s *cost function* and its *conditional factor demand correspondence*. As with the utility maximization and expenditure minimization problems in the theory of demand, there is a rich duality theory associated with the profit maximization and cost minimization problems.

Section 5.D analyzes in detail the geometry associated with cost and production relationships for the special but theoretically important case of a technology that produces a single output.

Aggregation theory is studied in Section 5.E. We show that aggregation on the supply side is simpler and more powerful than the corresponding theory for demand covered in Chapter 4.

Section 5.F constitutes an excursion into welfare economics. We define the concept of *efficient production* and study its relation to profit maximization. With some minor qualifications, we see that profit-maximizing production plans are efficient and that when suitable convexity properties hold, the converse is also true: An efficient plan is profit maximizing for an appropriately chosen vector of prices. This constitutes our first look at the important ideas of the *fundamental theorems of welfare economics*.

In Section 5.G, we point out that profit maximization does not have the same primitive status as preference maximization. Rigorously, it should be derived from the latter. We discuss this point and related issues.

In Appendix A, we study in more detail a particular, important case of production technologies: Those describable by means of linear constraints. It is known as the *linear activity model*.

5.B Production Sets

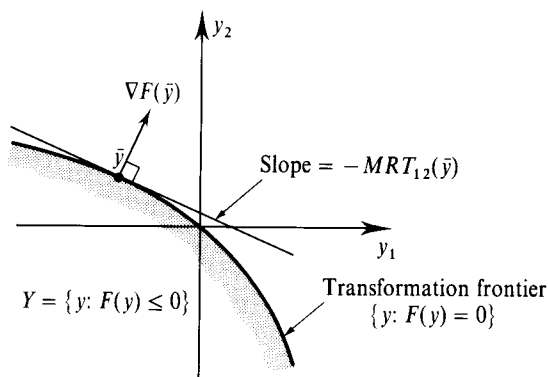
As in the previous chapters, we consider an economy with L commodities. A *production vector* (also known as an *input–output*, or *netput*, vector, or as a *production plan*) is a vector $y = (y_1, \dots, y_L) \in \mathbb{R}^L$ that describes the (net) outputs of the L commodities from a production process. We adopt the convention that positive numbers denote outputs and negative numbers denote inputs. Some elements of a production vector may be zero; this just means that the process has no net output of that commodity.

Example 5.B.1: Suppose that $L = 5$. Then $y = (-5, 2, -6, 3, 0)$ means that 2 and 3 units of goods 2 and 4, respectively, are produced, while 5 and 6 units of goods 1 and 3, respectively, are used. Good 5 is neither produced nor used as an input in this production vector. ■

To analyze the behavior of the firm, we need to start by identifying those production vectors that are technologically possible. The set of all production vectors that constitute feasible plans for the firm is known as the *production set* and is denoted by $Y \subset \mathbb{R}^L$. Any $y \in Y$ is possible; any $y \notin Y$ is not. The production set is taken as a primitive datum of the theory.

The set of feasible production plans is limited first and foremost by technological constraints. However, in any particular model, legal restrictions or prior contractual commitments may also contribute to the determination of the production set.

It is sometimes convenient to describe the production set Y using a function $F(\cdot)$, called the *transformation function*. The transformation function $F(\cdot)$ has the property that $Y = \{y \in \mathbb{R}^L: F(y) \leq 0\}$ and $F(y) = 0$ if and only if y is an element of the boundary of Y . The set of boundary points of Y , $\{y \in \mathbb{R}^L: F(y) = 0\}$, is known as the *transformation frontier*. Figure 5.B.1 presents a two-good example.

**Figure 5.B.1**

The production set and transformation frontier.

If $F(\cdot)$ is differentiable, and if the production vector $\bar{y}$ satisfies $F(\bar{y}) = 0$, then for any commodities ℓ and k , the ratio

$$MRT_{\ell k}(\bar{y}) = \frac{\partial F(\bar{y})/\partial y_{\ell}}{\partial F(\bar{y})/\partial y_k}$$

is called the *marginal rate of transformation (MRT) of good ℓ for good k at $\bar{y}$* .¹ The marginal rate of transformation is a measure of how much the (net) output of good k can increase if the firm decreases the (net) output of good ℓ by one marginal unit. Indeed, from $F(\bar{y}) = 0$, we get

$$\frac{\partial F(\bar{y})}{\partial y_k} dy_k + \frac{\partial F(\bar{y})}{\partial y_{\ell}} dy_{\ell} = 0,$$

and therefore the slope of the transformation frontier at $\bar{y}$ in Figure 5.B.1 is precisely $-MRT_{12}(\bar{y})$.

Technologies with Distinct Inputs and Outputs

In many actual production processes, the set of goods that can be outputs is distinct from the set that can be inputs. In this case, it is sometimes convenient to notationally distinguish the firm's inputs and outputs. We could, for example, let $q = (q_1, \dots, q_M) \geq 0$ denote the production levels of the firm's M outputs and $z = (z_1, \dots, z_{L-M}) \geq 0$ denote the amounts of the firm's $L - M$ inputs, with the convention that the amount of input z_{ℓ} used is now measured as a *nonnegative* number (as a matter of notation, we count all goods not actually used in the process as inputs).

One of the most frequently encountered production models is that in which there is a single output. A single-output technology is commonly described by means of a *production function* $f(z)$ that gives the maximum amount q of output that can be produced using input amounts $(z_1, \dots, z_{L-1}) \geq 0$. For example, if the output is good L , then (assuming that output can be disposed of at no cost) the production function $f(\cdot)$ gives rise to the production set:

$$Y = \{(-z_1, \dots, -z_{L-1}, q) : q - f(z_1, \dots, z_{L-1}) \leq 0 \text{ and } (z_1, \dots, z_{L-1}) \geq 0\}.$$

Holding the level of output fixed, we can define the *marginal rate of technical*

1. As in Chapter 3, in computing ratios such as this, we always assume that $\partial F(\bar{y})/\partial y_k \neq 0$.

substitution (MRTS) of input ℓ for input k at $\bar{z}$ as

$$MRTS_{\ell k}(\bar{z}) = \frac{\partial f(\bar{z})/\partial z_{\ell}}{\partial f(\bar{z})/\partial z_k}.$$

The number $MRTS_{\ell k}(\bar{z})$ measures the additional amount of input k that must be used to keep output at level $\bar{q} = f(\bar{z})$ when the amount of input ℓ is decreased marginally. It is the production theory analog to the consumer's marginal rate of substitution. In consumer theory, we look at the trade-off between commodities that keeps utility constant, here, we examine the trade-off between inputs that keeps the amount of output constant. Note that $MRTS_{\ell k}$ is simply a renaming of the marginal rate of transformation of input ℓ for input k in the special case of a single-output, many-input technology.

Example 5.B.2: The Cobb–Douglas Production Function The Cobb–Douglas production function with two inputs is given by $f(z_1, z_2) = z_1^{\alpha} z_2^{\beta}$, where $\alpha \geq 0$ and $\beta \geq 0$. The marginal rate of technical substitution between the two inputs at $z = (z_1, z_2)$ is $MRTS_{12}(z) = \alpha z_2 / \beta z_1$. ■

Properties of Production Sets

We now introduce and discuss a fairly exhaustive list of commonly assumed properties of production sets. The appropriateness of each of these assumptions depends on the particular circumstances (indeed, some of them are mutually exclusive).²

(i) *Y is nonempty.* This assumption simply says that the firm has something it can plan to do. Otherwise, there is no need to study the behavior of the firm in question.

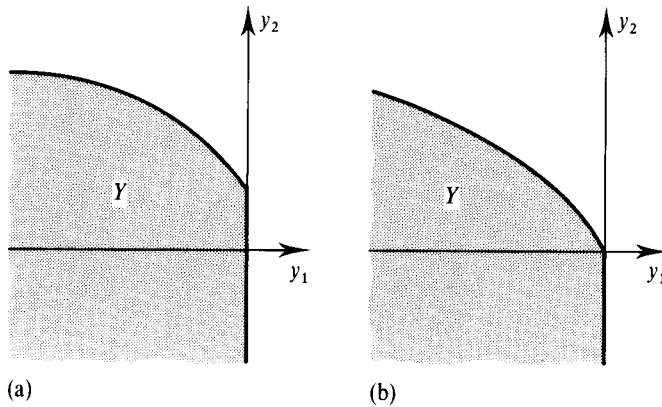
(ii) *Y is closed.* The set Y includes its boundary. Thus, the limit of a sequence of technologically feasible input–output vectors is also feasible; in symbols, $y^n \rightarrow y$ and $y^n \in Y$ imply $y \in Y$. This condition should be thought of as primarily technical.³

(iii) *No free lunch.* Suppose that $y \in Y$ and $y \geq 0$, so that the vector y does not use any inputs. The no-free-lunch property is satisfied if this production vector cannot produce output either. That is, whenever $y \in Y$ and $y \geq 0$, then $y = 0$; it is not possible to produce something from nothing. Geometrically, $Y \cap \mathbb{R}_+^L \subset \{0\}$. For $L = 2$, Figure 5.B.2(a) depicts a set that violates the no-free-lunch property, the set in Figure 5.B.2(b) satisfies it.

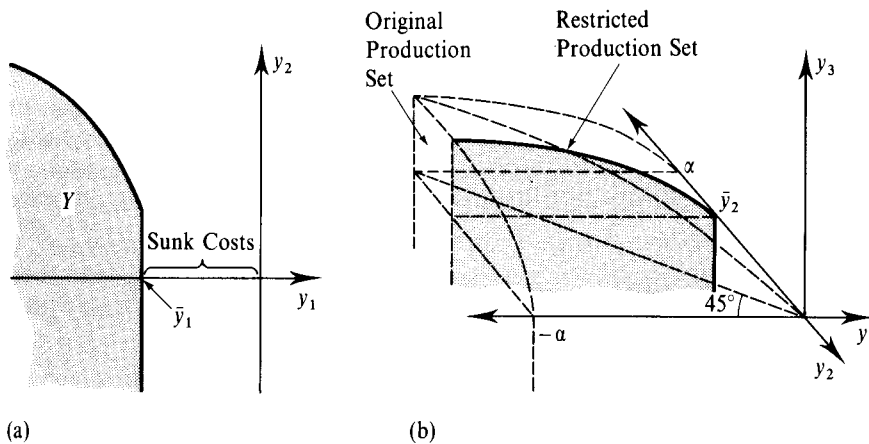
(iv) *Possibility of inaction* This property says that $0 \in Y$: Complete shutdown is possible. Both sets in Figure 5.B.2, for example, satisfy this property. The point in time at which production possibilities are being analyzed is often important for the validity of this assumption. If we are contemplating a firm that could access a set of technological possibilities but that has not yet been organized, then inaction is clearly

2. For further discussion of these properties, see Koopmans (1957) and Chapter 3 of Debreu (1959).

3. Nonetheless, we show in Exercise 5.B.4 that there is an important case of economic interest when it raises difficulties.

**Figure 5.B.2**

The no free lunch property.
 (a) Violates no free lunch.
 (b) Satisfies no free lunch.

**Figure 5.B.3**

Two production sets with sunk costs.
 (a) A minimal level of expenditure committed.
 (b) One kind of input fixed.

possible. But if some production decisions have already been made, or if irrevocable contracts for the delivery of some inputs have been signed, inaction is not possible. In that case, we say that some costs are *sunk*. Figure 5.B.3 depicts two examples. The production set in Figure 5.B.3(a) represents the *interim* production possibilities arising when the firm is already committed to use at least $-\bar{y}_1$ units of good 1 (perhaps because it has already signed a contract for the purchase of this amount); that is, the set is a *restricted production set* that reflects the firm's remaining choices from some original production set Y like the ones in Figure 5.B.2. In Figure 5.B.3(b), we have a second example of sunk costs. For a case with one output (good 3) and two inputs (goods 1 and 2), the figure illustrates the restricted production set arising when the level of the second input has been irrevocably set at $\bar{y}_2 < 0$ [here, in contrast with Figure 5.B.3(a), increases in the use of the input are impossible].

(v) *Free disposal*. The property of free disposal holds if the absorption of any additional amounts of inputs without any reduction in output is always possible. That is, if $y \in Y$ and $y' \leq y$ (so that y' produces at most the same amount of outputs using at least the same amount of inputs), then $y' \in Y$. More succinctly, $Y - \mathbb{R}_+^L \subset Y$ (see Figure 5.B.4). The interpretation is that the extra amount of inputs (or outputs) can be disposed of or eliminated at no cost.

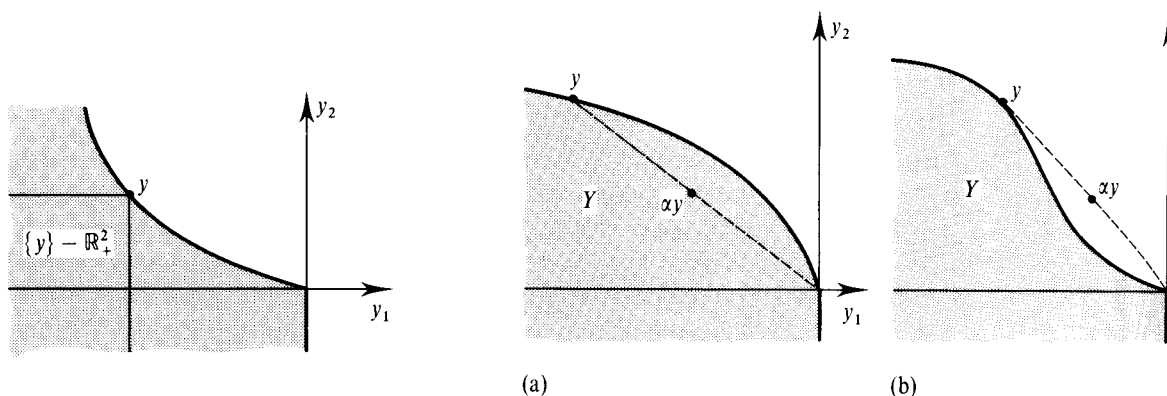


Figure 5.B.4 (left)
The free disposal property.

Figure 5.B.5 (right)
The nonincreasing returns to scale property.
(a) Nonincreasing returns satisfied.
(b) Nonincreasing returns violated.

(vi) *Irreversibility*. Suppose that $y \in Y$ and $y \neq 0$. Then irreversibility says that $-y \notin Y$. In words, it is impossible to reverse a technologically possible production vector to transform an amount of output into the same amount of input that was used to generate it. If, for example, the description of a commodity includes the time of its availability, then irreversibility follows from the requirement that inputs be used before outputs emerge.

Exercise 5.B.1: Draw two production sets: one that violates irreversibility and one that satisfies this property.

(vii) *Nonincreasing returns to scale*. The production technology Y exhibits nonincreasing returns to scale if for any $y \in Y$, we have $\alpha y \in Y$ for all scalars $\alpha \in [0, 1]$. In words, any feasible input–output vector can be scaled down (see Figure 5.B.5). Note that nonincreasing returns to scale imply that inaction is possible [property (iv)].

(viii) *Nondecreasing returns to scale*. In contrast with the previous case, the production process exhibits nondecreasing returns to scale if for any $y \in Y$, we have $\alpha y \in Y$ for any scale $\alpha \geq 1$. In words, any feasible input–output vector can be scaled up. Figure 5.B.6(a) presents a typical example; in the figure, units of output (good 2) can be produced at a constant cost of input (good 1) except that in order to produce at all, a fixed setup cost is required. It does not matter for the existence of nondecreasing returns if this fixed cost is sunk [as in Figure 5.B.6(b)] or not [as in Figure 5.B.6(a), where inaction is possible].

(ix) *Constant returns to scale*. This property is the conjunction of properties (vii) and (viii). The production set Y exhibits constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for any scalar $\alpha \geq 0$. Geometrically, Y is a *cone* (see Figure 5.B.7).

For single-output technologies, properties of the production set translate readily into properties of the production function $f(\cdot)$. Consider Exercise 5.B.2 and Example 5.B.3.

Exercise 5.B.2: Suppose that $f(\cdot)$ is the production function associated with a single-output technology, and let Y be the production set of this technology. Show that Y satisfies constant returns to scale if and only if $f(\cdot)$ is homogeneous of degree one.

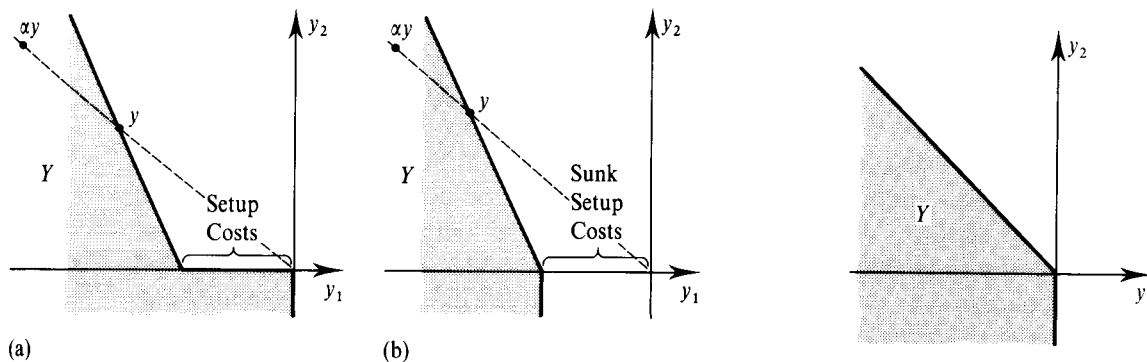


Figure 5.B.6 (left)
The nondecreasing returns to scale property.

Figure 5.B.7 (right)
A technology satisfying the constant returns to scale property.

Example 5.B.3: Returns to Scale with the Cobb–Douglas Production Function: For the Cobb–Douglas production function introduced in Example 5.B.2, $f(2z_1, 2z_2) = 2^{\alpha+\beta} z_1^\alpha z_2^\beta = 2^{\alpha+\beta} f(z_1, z_2)$. Thus, when $\alpha + \beta = 1$, we have constant returns to scale; when $\alpha + \beta < 1$, we have decreasing returns to scale; and when $\alpha + \beta > 1$, we have increasing returns to scale. ■

(x) *Additivity (or free entry).* Suppose that $y \in Y$ and $y' \in Y$. The additivity property requires that $y + y' \in Y$. More succinctly, $Y + Y \subset Y$. This implies, for example, that $ky \in Y$ for any positive integer k . In Figure 5.B.8, we see an example where Y is additive. Note that in this example, output is available only in integer amounts (perhaps because of indivisibilities). The economic interpretation of the additivity condition is that if y and y' are both possible, then one can set up two plants that do not interfere with each other and carry out production plans y and y' independently. The result is then the production vector $y + y'$.

Additivity is also related to the idea of entry. If $y \in Y$ is being produced by a firm and another firm enters and produces $y' \in Y$, then the net result is the vector $y + y'$. Hence, the *aggregate production set* (the production set describing feasible production plans for the economy as a whole) must satisfy additivity whenever unrestricted entry, or (as it is called in the literature) *free entry*, is possible.

(xi) *Convexity.* This is one of the fundamental assumptions of microeconomics. It postulates that the production set Y is convex. That is, if $y, y' \in Y$ and $\alpha \in [0, 1]$, then $\alpha y + (1 - \alpha)y' \in Y$. For example, Y is convex in Figure 5.B.5(a) but is not convex in Figure 5.B.5(b).

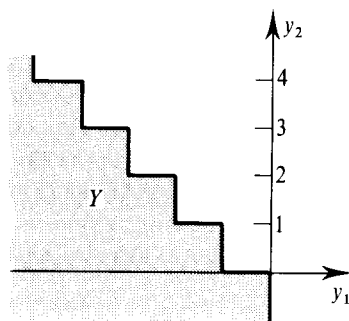


Figure 5.B.8
A production set satisfying the additivity property.

The convexity assumption can be interpreted as incorporating two ideas about production possibilities. The first is nonincreasing returns. In particular, if inaction is possible (i.e., if $0 \in Y$), then convexity implies that Y has nonincreasing returns to scale. To see this, note that for any $\alpha \in [0, 1]$, we can write $\alpha y = \alpha y + (1 - \alpha)0$. Hence, if $y \in Y$ and $0 \in Y$, convexity implies that $\alpha y \in Y$. Second, convexity captures the idea that “unbalanced” input combinations are not more productive than balanced ones (or, symmetrically, that “unbalanced” output combinations are not least costly to produce than balanced ones). In particular, if production plans y and y' produce exactly the same amount of output but use different input combinations, then a production vector that uses a level of each input that is the average of the levels used in these two plans can do at least as well as either y or y' .

Exercise 5.B.3 illustrates these two ideas for the case of a single-output technology.

Exercise 5.B.3: Show that for a single-output technology, Y is convex if and only if the production function $f(z)$ is concave.

(xii) Y is a convex cone. This is the conjunction of the convexity (xi) and constant returns to scale (ix) properties. Formally, Y is a convex cone if for any production vector $y, y' \in Y$ and constants $\alpha \geq 0$ and $\beta \geq 0$, we have $\alpha y + \beta y' \in Y$. The production set depicted in Figure 5.B.7 is a convex cone.

An important fact is given in Proposition 5.B.1.

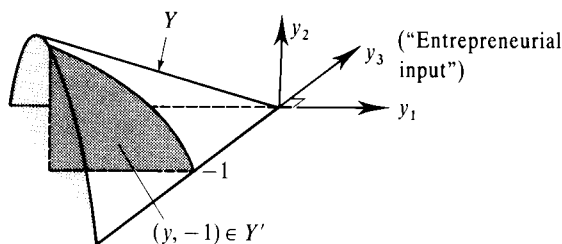
Proposition 5.B.1: The production set Y is additive and satisfies the nonincreasing returns condition if and only if it is a convex cone.

Proof: The definition of a convex cone directly implies the nonincreasing returns and additivity properties. Conversely, we want to show that if nonincreasing returns and additivity hold, then for any $y, y' \in Y$ and any $\alpha > 0$, and $\beta > 0$, we have $\alpha y + \beta y' \in Y$. To this effect, let k be any integer such that $k > \text{Max}\{\alpha, \beta\}$. By additivity, $ky \in Y$ and $ky' \in Y$. Since $(\alpha/k) < 1$ and $\alpha y = (\alpha/k)ky$, the nonincreasing returns condition implies that $\alpha y \in Y$. Similarly, $\beta y' \in Y$. Finally, again by additivity, $\alpha y + \beta y' \in Y$. ■

Proposition 5.B.1 provides a justification for the convexity assumption in production. Informally, we could say that if feasible input–output combinations can always be scaled down, and if the simultaneous operation of several technologies without mutual interference is always possible, then, in particular, convexity obtains. (See Appendix A of Chapter 11 for several examples in which there is mutual interference and, as a consequence, convexity does not arise.)

It is important not to lose sight of the fact that the production set describes technology, not limits on resources. It can be argued that if all inputs (including, say, entrepreneurial inputs) are explicitly accounted for, then it should always be possible to replicate production. After all, we are not saying that doubling output is actually feasible, only that in principle it would be possible if *all* inputs (however esoteric, be they marketed or not) were doubled. In this view, which originated with Marshall and has been much emphasized by McKenzie (1959), decreasing returns must reflect the scarcity of an underlying, unlisted input of production. For this reason, some economists believe that among models with convex technologies the constant returns model is the most fundamental. Proposition 5.B.2 makes this idea precise.

Proposition 5.B.2: For any convex production set $Y \subset \mathbb{R}^L$ with $0 \in Y$, there is a constant returns, convex production set $Y' \subset \mathbb{R}^{L+1}$ such that $Y = \{y \in \mathbb{R}^L : (y, -1) \in Y'\}$.

**Figure 5.B.9**

A constant returns production set with an “entrepreneurial factor.”

Proof: Simply let $Y' = \{y' \in \mathbb{R}^{L+1}; y' = \alpha(y, -1) \text{ for some } y \in Y \text{ and } \alpha \geq 0\}$. (See Figure 5.B.9.) ■

The additional input included in the extended production set (good $L + 1$) can be called the “entrepreneurial factor.” (The justification for this can be seen in Exercise 5.C.12; in a competitive environment, the return to this entrepreneurial factor is precisely the firm’s profit.) In essence, the implication of Proposition 5.B.2 is that in a competitive, convex setting, there may be little loss of conceptual generality in limiting ourselves to constant returns technologies.

5.C Profit Maximization and Cost Minimization

In this section, we begin our study of the market behavior of the firm. In parallel to our study of consumer demand, we assume that there is a vector of prices quoted for the L goods, denoted by $p = (p_1, \dots, p_L) \gg 0$, and that these prices are independent of the production plans of the firm (the *price-taking assumption*).

We assume throughout this chapter that the firm’s objective is to maximize its profit. (It is quite legitimate to ask why this should be so, and we will offer a brief discussion of the issue in Section 5.G.) Moreover, we always assume that the firm’s production set Y satisfies the properties of *nonemptiness*, *closedness*, and *free disposal* (see Section 5.B).

The Profit Maximization Problem

Given a price vector $p \gg 0$ and a production vector $y \in \mathbb{R}^L$, the profit generated by implementing y is $p \cdot y = \sum_{\ell=1}^L p_{\ell} y_{\ell}$. By the sign convention, this is precisely the total revenue minus the total cost. Given the technological constraints represented by its production set Y , the firm’s *profit maximization problem* (PMP) is then

$$\begin{aligned} \text{Max}_y \quad & p \cdot y \\ \text{s.t.} \quad & y \in Y. \end{aligned} \tag{PMP}$$

Using a transformation function to describe Y , $F(\cdot)$, we can equivalently state the PMP as

$$\begin{aligned} \text{Max}_y \quad & p \cdot y \\ \text{s.t.} \quad & F(y) \leq 0. \end{aligned}$$